

probability introduced through sets and relative frequency

- Experiment:- a ***random experiment*** is an action or process that leads to one of several possible outcomes

Experiment	Outcomes
Flip a coin	Heads, Tails
Exam Marks	Numbers: 0, 1, 2, ..., 100
Assembly Time	$t > 0$ seconds
Course Grades	F, D, C, B, A, A+

Sample Space

- List: “Called the Sample Space”
- Outcomes: “Called the Simple Events”

This list must be ***exhaustive***, i.e. ALL possible outcomes included.

- Die roll {1,2,3,4,5} Die roll {1,2,3,4,5,6}
- The list must be ***mutually exclusive***, i.e. no two outcomes can occur at the same time:
- Die roll {odd number or even number}
- Die roll { number less than 4 or even number}

Sample Space

- A list of exhaustive [don't leave anything out] and mutually exclusive outcomes [impossible for 2 different events to occur in the same experiment] is called a ***sample space*** and is denoted by S .
- The outcomes are denoted by $O_1, O_2, \dots, O_k$
- Using notation from set theory, we can represent the sample space and its outcomes as:
 - $S = \{O_1, O_2, \dots, O_k\}$

- Given a sample space $S = \{O_1, O_2, \dots, O_k\}$, the ***probabilities*** assigned to the outcome must satisfy these requirements:

(1) The probability of any outcome is between 0 and 1

- i.e. $0 \leq P(O_i) \leq 1$ for each i , and

(2) The sum of the probabilities of all the outcomes equals 1

- i.e. $P(O_1) + P(O_2) + \dots + P(O_k) = 1$

$$\sum_{i=1}^k P(O_i) = 1$$

Relative Frequency

Random experiment with sample space S . we shall assign non-negative number called probability to each event in the sample space.

Let A be a particular event in S . then “the probability of event A ” is denoted by $P(A)$.

Suppose that the random experiment is repeated n times, if the event A occurs n_A times, then the probability of event A is defined as “Relative frequency “

- **Relative Frequency Definition:** The probability of an
- event A is defined as

$$P(A) = \lim_{n \rightarrow \infty} \frac{n_A}{n}$$

Axioms of Probability

For any event A , we assign a number $P(A)$, called the probability of the event A . This number satisfies the following three conditions that act the axioms of probability.

- (i) $P(A) \geq 0$ (Probability is a nonnegative number)
- (ii) $P(\Omega) = 1$ (Probability of the whole set is unity)
- (iii) If $A \cap B = \phi$, then $P(A \cup B) = P(A) + P(B)$.

(Note that (iii) states that if A and B are mutually exclusive (M.E.) events, the probability of their union is the sum of their probabilities.)

Events

- The ***probability of an event*** is the **sum** of the probabilities of the simple events that constitute the event.
- E.g. (assuming a fair die) $S = \{1, 2, 3, 4, 5, 6\}$ and
- $P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = 1/6$
- Then:
- $P(\text{EVEN}) = P(2) + P(4) + P(6) = 1/6 + 1/6 + 1/6 = 3/6 = 1/2$

Conditional Probability

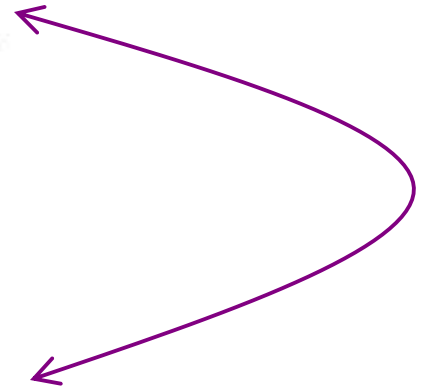
- **Conditional probability** is used to determine how two events are related; that is, we can determine the probability of one event **given** the occurrence of another related event.
- Experiment: random select one student in class.
- $P(\text{randomly selected student is male}) =$
- $P(\text{randomly selected student is male/student is on 3}^{\text{rd}} \text{ row}) =$
- Conditional probabilities are written as **$P(A \mid B)$** and read as “the probability of *A* *given* *B*” and is calculated as

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$

- $P(A \text{ and } B) = P(A) * P(B/A) = P(B) * P(A/B)$ both are true
- Keep this in mind!

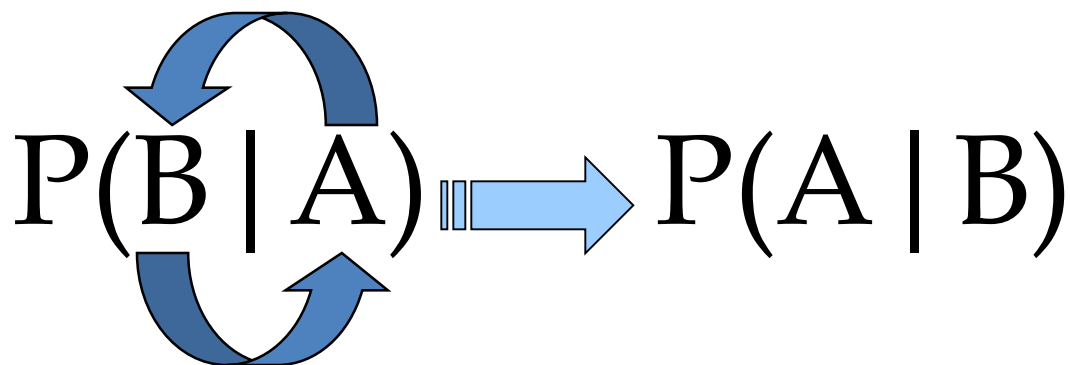
$$P(A | B) = \frac{P(A \text{ and } B)}{P(B)}$$

$$P(B | A) = \frac{P(A \text{ and } B)}{P(A)}$$



Bayes' Law

- Bayes' Law is named for Thomas Bayes, an eighteenth century mathematician.
- In its most basic form, if we know $P(B | A)$,
- we can apply Bayes' Law to determine $P(A | B)$



- The probabilities $P(A)$ and $P(A^C)$ are called ***prior probabilities*** because they are determined **prior** to the decision about taking the preparatory course.
- The conditional probability $P(A \mid B)$ is called a ***posterior probability*** (or revised probability), because the prior probability is revised **after** the decision about taking the preparatory course.

Total probability theorem

- Take events A_i for $i = 1$ to k to be:
 - Mutually exclusive: $A_i \cap A_j = \emptyset$ for all i, j
 - Exhaustive: $A_1 \cup \dots \cup A_k = S$

For any event B on S

$$p(B) = p(B|A_1)p(A_1) + \dots + p(B|A_k)p(A_k)$$

$$p(B) = \sum_{i=1}^k p(B|A_i)p(A_i)$$

Bayes theorem follows

$$p(A_j|B) = \frac{p(A_j \cap B)}{p(B)} = \frac{p(B|A_j) \cdot p(A_j)}{\sum_{i=1}^k p(B|A_i)p(A_i)}$$

Independence

- Do A and B depend on one another?
 - Yes! B more likely to be true if A.
 - A should be more likely if B.

- If Independent

$$p(A \cap B) = p(A) \cdot p(B)$$

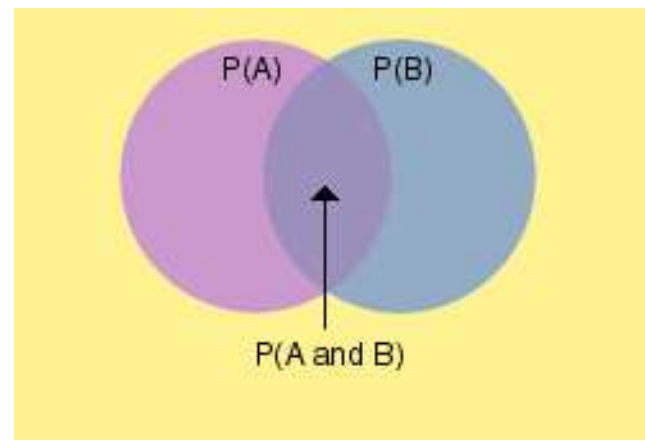
$$p(A|B) = p(A) \quad p(B|A) = p(B)$$

- If Dependent

$$p(A \cap B) \neq p(A) \cdot p(B)$$

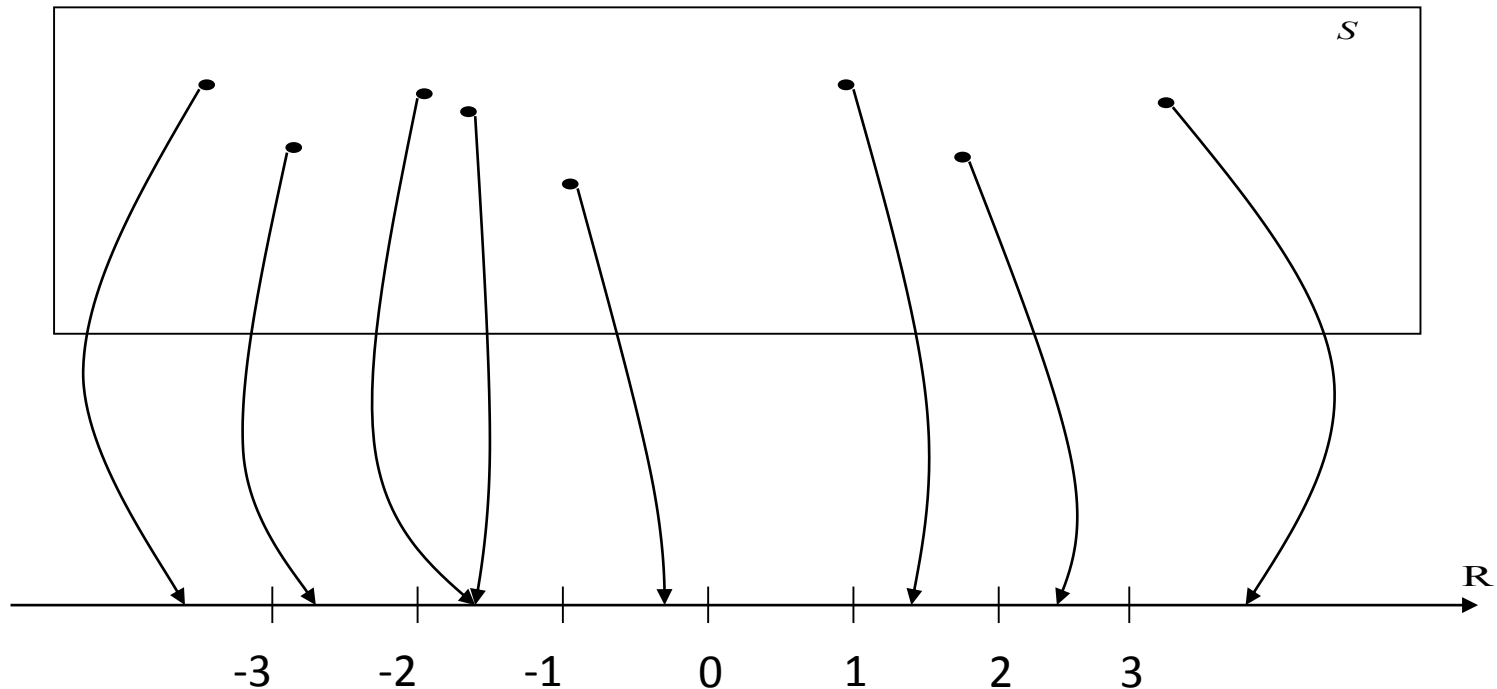
$$p(A \cup B) = p(A) + p(B) - p(A \cap B)$$

$$p(A \cap B) = p(B|A) \cdot p(A)$$



Random variable

- Random variable
 - A numerical value to each outcome of a particular experiment



- Example 1 : Machine Breakdowns
 - Sample space : $S = \{electrical, mechanical, misuse\}$
 - Each of these failures may be associated with a repair cost
 - State space : $\{50, 200, 350\}$
 - Cost is a random variable : 50, 200, and 350
- Probability Mass Function (p.m.f.)
 - A set of probability value assigned to each of the values taken by the discrete random variable x_i
 - $0 \leq p_i \leq 1$ and $\sum_i p_i = 1$
 - Probability : $P(X = x_i) = p_i$

Continuous and Discrete random variables

- **Discrete** random variables have a countable number of outcomes
 - Examples: Dead/alive, treatment/placebo, dice, counts, etc.
- **Continuous** random variables have an infinite continuum of possible values.
 - Examples: blood pressure, weight, the speed of a car, the real numbers from 1 to 6.

- **Distribution function:**

$$F_X(x) = P(X \leq x), \quad -\infty < x < \infty$$

- If $F_X(x)$ is a continuous function of x , then X is a continuous random variable.

- $F_X(x)$: discrete in $x \rightarrow$ Discrete rv's

- $F_X(x)$: piecewise continuous $\rightarrow$ Mixed rv's

- **PROPERTIES:**

- $0 \leq F_X(x) \leq 1, \quad -\infty < x < \infty$

- $F_X(x)$: monotonically increasing func. of x

- $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow \infty} F_X(x) = 1$

Probability Density Function (pdf)

- X : continuous rv, then, $f(x) = \frac{dF(x)}{dx}$ is the *pdf* of X .

$$CDF \longleftrightarrow pdf$$

$$P(X \leq x) = F(x) = \int_{-\infty}^x f(u) du, \quad -\infty < x < \infty$$

$$P(X \in (a, b]) = P(a < X \leq b) = \int_a^b f_X(u) du.$$

- **pdf properties:**

1. $f(x) \geq 0$ for all x .

2. $\int_{-\infty}^{\infty} f(x) dx = 1.$ $F(t) = \int_{-\infty}^t f(x) dx$
 $= \int_0^t f(x) dx \quad ,$

Binomial

- Suppose that the probability of success is p

- What is the probability of failure?

$$q = 1 - p$$

- Examples

- Toss of a coin ($S = \text{head}$): $p = 0.5 \Rightarrow q = 0.5$

- Roll of a die ($S = 1$): $p = 0.1667 \Rightarrow q = 0.8333$

- Fertility of a chicken egg ($S = \text{fertile}$): $p = 0.8 \Rightarrow q = 0.2$

binomial

- Imagine that a trial is repeated n times
- Examples
 - A coin is tossed 5 times
 - A die is rolled 25 times
 - 50 chicken eggs are examined
- Assume p remains constant from trial to trial and that the trials are statistically independent of each other
- Example
 - What is the probability of obtaining 2 heads from a coin that was tossed 5 times?

$$P(HHTTT) = (1/2)^5 = 1/32$$

Poisson

- When there is a large number of trials, but a small probability of success, binomial calculation becomes impractical
 - Example: Number of deaths from horse kicks in the Army in different years
- The mean number of successes from n trials is $\mu = np$
 - Example: 64 deaths in 20 years from thousands of soldiers

If we substitute μ/n for p , and let n tend to infinity, the binomial distribution becomes the Poisson distribution:

$$P(x) = \frac{e^{-\mu}\mu^x}{x!}$$

poisson

- Poisson distribution is applied where random events in space or time are expected to occur
- Deviation from Poisson distribution may indicate some degree of non-randomness in the events under study
- Investigation of cause may be of interest

Exponential Distribution

The random variable X that equals the distance between successive counts of a Poisson process with mean $\lambda > 0$ is an **exponential random variable** with parameter λ . The probability density function of X is

$$f(x) = \lambda e^{-\lambda x} \quad \text{for } 0 \leq x < \infty \quad (4-14)$$

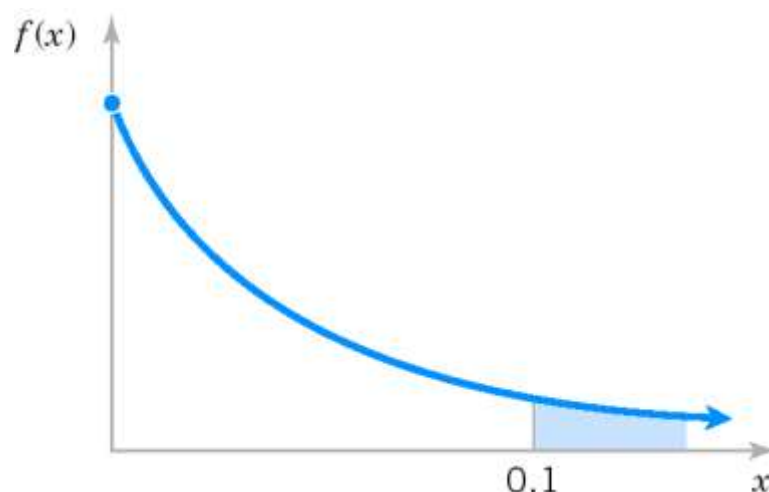
If the random variable X has an exponential distribution with parameter λ ,

$$\mu = E(X) = \frac{1}{\lambda} \quad \text{and} \quad \sigma^2 = V(X) = \frac{1}{\lambda^2} \quad (4-15)$$

In a large corporate computer network, user log-ons to the system can be modeled as a Poisson process with a mean of 25 log-ons per hour. What is the probability that there are no log-ons in an interval of 6 minutes?

Let X denote the time in hours from the start of the interval until the first log-on. Then, X has an exponential distribution with $\lambda = 25$ log-ons per hour. We are interested in the probability that X exceeds 6 minutes. Because λ is given in log-ons per hour, we express all time units in hours. That is, 6 minutes = 0.1 hour. The probability requested is shown as the shaded area under the probability density function in Fig. 4-23. Therefore,

$$P(X > 0.1) = \int_{0.1}^{\infty} 25e^{-25x} dx = e^{-25(0.1)} = 0.082$$



Also, the cumulative distribution function can be used to obtain the same result as follows:

$$P(X > 0.1) = 1 - F(0.1) = e^{-25(0.1)}$$

An identical answer is obtained by expressing the mean number of log-ons as 0.417 log-ons per minute and computing the probability that the time until the next log-on exceeds 6 minutes. Try it.

What is the probability that the time until the next log-on is between 2 and 3 minutes? Upon converting all units to hours,

$$P(0.033 < X < 0.05) = \int_{0.033}^{0.05} 25e^{-25x} dx = -e^{-25x} \Big|_{0.033}^{0.05} = 0.152$$

An alternative solution is

$$P(0.033 < X < 0.05) = F(0.05) - F(0.033) = 0.152$$

Determine the interval of time such that the probability that no log-on occurs in the interval is 0.90. The question asks for the length of time x such that $P(X > x) = 0.90$. Now,

$$P(X > x) = e^{-25x} = 0.90$$

Take the (natural) log of both sides to obtain $-25x = \ln(0.90) = -0.1054$. Therefore,

$$x = 0.00421 \text{ hour} = 0.25 \text{ minute}$$

Furthermore, the mean time until the next log-on is

$$\mu = 1/25 = 0.04 \text{ hour} = 2.4 \text{ minutes}$$

The standard deviation of the time until the next log-on is

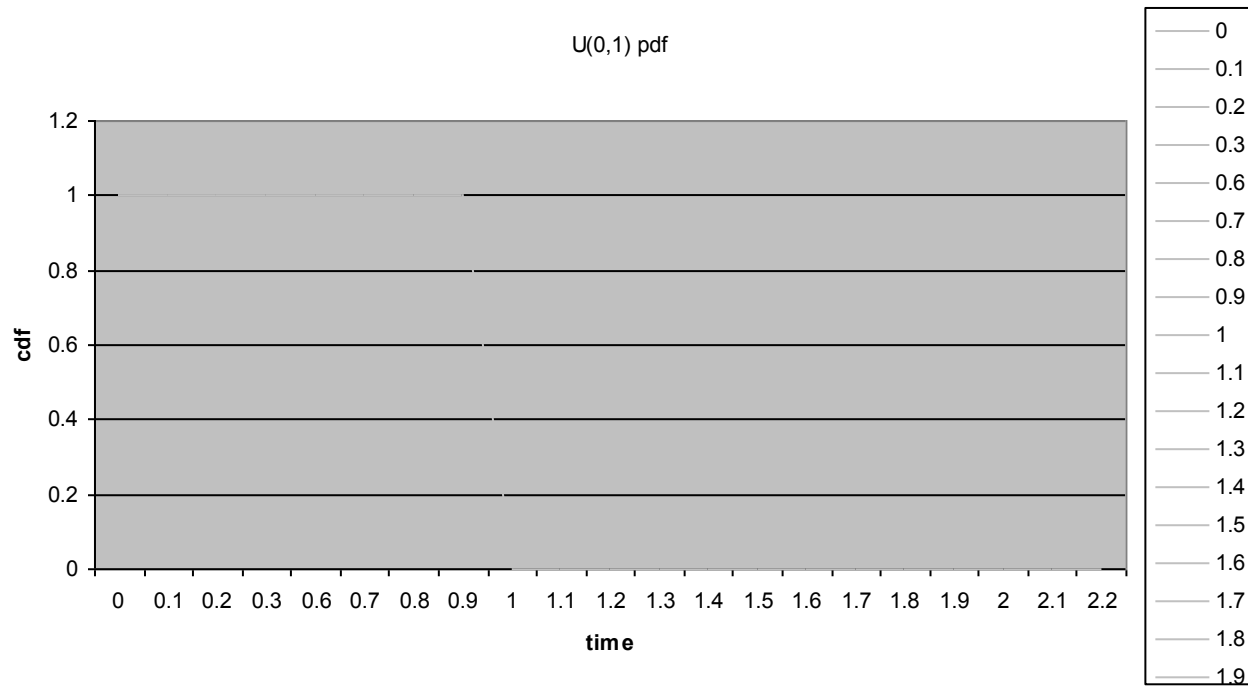
$$\sigma = 1/25 \text{ hours} = 2.4 \text{ minutes}$$

Uniform

All (pseudo) random generators generate random deviates of $U(0,1)$ distribution; that is, if you generate a large number of random variables and plot their empirical distribution function, it will approach this distribution in the limit.

$U(a,b) \rightarrow$ pdf constant over the (a,b) interval and CDF is the ramp function

$$f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$$



Uniform distribution

$$F(x) = \begin{cases} 0, & x < a, \\ \frac{x-a}{b-a}, & a \leq x < b \\ 1, & x \geq b. \end{cases}$$

Gaussian (Normal) Distribution

- Bell shaped pdf – intuitively pleasing!
- Central Limit Theorem: *mean of a large number of mutually independent rv's (having arbitrary distributions) starts following Normal distribution as $n \rightarrow$*

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$$

- μ : mean, σ : std. deviation, σ^2 : variance ($N(\mu, \sigma^2)$)
- μ and σ completely describe the statistics. This is significant in statistical estimation/signal processing/communication theory etc.

- $N(0,1)$ is called normalized Gaussian.
- $N(0,1)$ is symmetric i.e.
 - $f(x)=f(-x)$
 - $F(z) = 1-F(-z)$.
- Failure rate $h(t)$ follows IFR behavior.
 - Hence, $N()$ is suitable for modeling long-term wear or aging related failure phenomena

Exponential Distribution

$$f(t) = \sum_{i=1}^k \alpha_i \lambda_i e^{-\lambda_i t}, \quad t > 0, \quad \lambda_i > 0, \quad \alpha_i > 0, \quad \sum_{i=1}^k \alpha_i$$

$$F(t) = \sum_i \alpha_i (1 - e^{-\lambda_i t}), \quad t \geq 0$$

$$h(t) = \frac{\sum_i \alpha_i \lambda_i e^{-\lambda_i t}}{\sum_i \alpha_i \lambda_i e^{-\lambda_i t}}, \quad t \geq 0$$

Conditional Distributions

- The conditional distribution of Y *given* $X=1$ is:
- While marginal distributions are obtained from the bivariate by summing, conditional distributions are obtained by “making a cut” through the bivariate distribution

The Expectation of a Random Variable

Expectation of a discrete random variable with p.m.f

$$P(X = x_i) = p_i$$

$$E(X) = \sum_i p_i x_i$$

Expectation of a continuous random variable with p.d.f $f(x)$

$$E(X) = \int_{\text{state space}} x f(x) dx$$

expectation of X = mean of X = average of X

$$E[X] = \bar{X} = \int_{-\infty}^{\infty} x f_X(x) dx \quad \text{continuous r.v.}$$

$$E[X] = \bar{X} = \sum_{i=1}^N x_i P(x_i) \quad \text{discrete r.v.}$$

$$f_X(x+a) = f_X(-x+a), \forall x \Rightarrow E[X] = a$$

$$X \text{ r.v.} \Rightarrow Y=g(X) \text{ r.v.} \quad \text{Ex: } Y = g(X) = X^2$$

$$P(X=0) = P(X=-1) = P(X=1) = \frac{1}{3} \quad P(Y=0) = \frac{1}{3} \quad P(Y=1) = \frac{2}{3}$$

Expectation

expectation of a function of a r.v. X

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx \quad \text{continuous r.v.}$$

$$E[g(X)] = \sum_{i=1}^N g(x_i) P(x_i) \quad \text{discrete r.v.}$$

conditional expectation of a r.v. X

$$E[X|B] = \int_{-\infty}^{\infty} x f_X(x|B) dx \quad \text{continuous r.v.}$$

$$E[X|B] = \sum_{i=1}^N x_i P(x_i|B) \quad \text{discrete r.v.}$$

Ex: $B = \{X \leq b\}$

$$f_X(x|X \leq b) = \begin{cases} \frac{f_X(x)}{\int_{-\infty}^b f_X(x)dx}, & x < b \\ 0, & x \geq b \end{cases}$$

$$E[X|X \leq b] = \frac{\int_{-\infty}^b x f_X(x) dx}{\int_{-\infty}^b f_X(x) dx}$$

Moments

n -th moment of a r.v. X

$$m_n = E[X^n] = \int_{-\infty}^{\infty} x^n f_X(x) dx$$

continuous r.v.

$$m_n = E[X^n] = \sum_{i=1}^N x_i^n P(x_i)$$

discrete r.v.

$$m_0 = 1$$

$$m_1 = \bar{X}$$

properties of expectation:

$$(1) \quad E[c] = c \quad c \text{ -- constant}$$

$$(2) \quad E[ag(X) + bh(X)] = aE[g(X)] + bE[h(X)]$$

$$\text{PF: } E[c] = \int_{-\infty}^{\infty} cf_X(x)dx = c \int_{-\infty}^{\infty} f_X(x)dx = c$$

$$E[ag(X) + bh(X)] = \int_{-\infty}^{\infty} \{ag(x) + bh(x)\} f_X(x)dx$$

$$= a \int_{-\infty}^{\infty} g(x) f_X(x)dx + b \int_{-\infty}^{\infty} h(x) f_X(x)dx = aE[g(X)] + bE[h(X)]$$

variance of a r.v. X

$$\begin{aligned}\sigma_X^2 &= \mu_2 = E[(X - \bar{X})^2] = E[X^2 - 2\bar{X}X + \bar{X}^2] \\ &= E[X^2] - 2\bar{X}E[X] + \bar{X}^2 = m_2 - m_1^2\end{aligned}$$

standard deviation of a r.v. $X = \sigma_X (\geq 0)$

skewness of a r.v. $X = \frac{\mu_3}{\sigma_X^3}$

Ex 3.2-1 & Ex3.2-2: $f_X(x)$ symmetric about $x = \bar{X} \Rightarrow \mu_3 = 0$

exponential r.v.

$$f_X(x) = \begin{cases} \frac{1}{b} e^{-\frac{x-a}{b}}, & x > a \\ 0, & x < a \end{cases}$$

$$m_1 = E[X] = \int_a^\infty x \frac{1}{b} e^{-\frac{x-a}{b}} dx = a + b$$

$$m_2 = E[X^2] = \int_a^\infty x^2 \frac{1}{b} e^{-\frac{x-a}{b}} dx = (a + b)^2 + b^2$$

$$\sigma_X^2 = \mu_2 = m_2 - m_1^2 = b^2$$

$$m_3 = E[X^3] = \int_a^\infty x^3 \frac{1}{b} e^{-\frac{x-a}{b}} dx = a^3 + 3a^2b + 6ab^2 + 6b^3$$

$$\begin{aligned} \mu_3 = E[(X - \bar{X})^3] &= E[X^3 - 3X^2\bar{X} + 3X\bar{X}^2 - \bar{X}^3] = m_3 - 3m_1m_2 + 3m_1^2m_1 - m_1^3 \\ &= a^3 + 3a^2b + 6ab^2 + 6b^3 - 3(a + b)\{(a + b)^2 + b^2\} + 2(a + b)^3 = 2b^3 \end{aligned}$$

$$\text{skewness of a r.v. } X = \frac{\mu_3}{\sigma_X^3} = \frac{2b^3}{b^3} = 2$$

Chebyshev's inequality $P[|X - \bar{X}| \geq \varepsilon] \leq \frac{\sigma_X^2}{\varepsilon^2}$

$$\begin{aligned} \sigma_X^2 &= \int_{-\infty}^{\infty} (x - \bar{X})^2 f_X(x) dx \geq \int_{|x - \bar{X}| \geq \varepsilon} (x - \bar{X})^2 f_X(x) dx \\ &\geq \varepsilon^2 \int_{|x - \bar{X}| \geq \varepsilon} f_X(x) dx = \varepsilon^2 P[|X - \bar{X}| \geq \varepsilon] \end{aligned}$$

Markov's inequality $P[X < 0] = 0 \Rightarrow P[X \geq a] \leq \frac{E[X]}{a}$

Ex 3.2-3: $P[|X - \bar{X}| \geq 3\sigma_X] \leq \frac{\sigma_X^2}{9\sigma_X^2} = \frac{1}{9}$

Characteristic function of r.v. X

$$\Phi_X(\omega) = E[e^{j\omega X}] = \int_{-\infty}^{\infty} f_X(x) e^{j\omega x} dx$$

$$f_X(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \Phi_X(\omega) e^{-j\omega x} d\omega$$

Fourier transform

$$|\Phi_X(\omega)| \leq \int_{-\infty}^{\infty} |f_X(x)| |e^{j\omega x}| dx \leq \int_{-\infty}^{\infty} f_X(x) dx = 1 = \Phi_X(0)$$

$$\left. \frac{d^n \Phi_X(\omega)}{d\omega^n} \right|_{\omega=0} = \int_{-\infty}^{\infty} f_X(x) j^n x^n e^{j\omega x} dx \Big|_{\omega=0} = j^n \int_{-\infty}^{\infty} f_X(x) x^n dx = j^n E[X^n]$$

$$m_n = (-j)^n \left. \frac{d^n \Phi_X(\omega)}{d\omega^n} \right|_{\omega=0}$$

Functions That Give Moments

Moment generating function of r.v. X

$$M_X(v) = E[e^{vX}] = \int_{-\infty}^{\infty} f_X(x) e^{vx} dx$$

$$\left. \frac{d^n M_X(v)}{dv^n} \right|_{v=0} = \int_{-\infty}^{\infty} f_X(x) x^n e^{vx} dx \Big|_{v=0} = \int_{-\infty}^{\infty} f_X(x) x^n dx = m_n$$

Ex 3.3-1 & Ex 3.3-2:

$$f_X(x) = \begin{cases} \frac{1}{b} e^{-\frac{x-a}{b}}, & x > a \\ 0, & x < a \end{cases}$$

$$\begin{aligned}\Phi_X(\omega) &= E[e^{j\omega X}] = \frac{1}{b} e^{\frac{a}{b}} \int_a^\infty e^{-(\frac{1}{b} - j\omega)x} dx = \frac{1}{b} e^{\frac{a}{b}} \frac{e^{-(\frac{1}{b} - j\omega)x}}{-(\frac{1}{b} - j\omega)} \bigg|_{x=a}^\infty \\ &= \frac{1}{b} e^{\frac{a}{b}} \frac{e^{-(\frac{1}{b} - j\omega)a}}{(\frac{1}{b} - j\omega)} = \frac{e^{j\omega a}}{1 - j\omega b}\end{aligned}$$

$$\frac{d\Phi_X(\omega)}{d\omega} = \frac{j a e^{j\omega a} (1 - j\omega b) + e^{j\omega a} j b}{(1 - j\omega b)^2}$$

$$M_X(v) = E[e^{vX}] = \frac{e^{va}}{1 - vb}$$

$$\frac{dM_X(v)}{dv} = \frac{a e^{va} (1 - vb) + e^{va} b}{(1 - vb)^2}$$

$$m_1 = (-j) \frac{d\Phi_X(\omega)}{d\omega} \bigg|_{\omega=0} = a + b$$

$$m_1 = \frac{dM_X(v)}{dv} \bigg|_{v=0} = a + b$$

Chernoff's inequality

Ex 3.3-3:

$$v > 0$$

$$\begin{aligned} P[X \geq a] &= \int_a^{\infty} f_X(x) dx = \int_{-\infty}^{\infty} f_X(x) u(x-a) dx \\ &\leq \int_{-\infty}^{\infty} f_X(x) e^{v(x-a)} dx = e^{-va} M_X(v) \end{aligned}$$

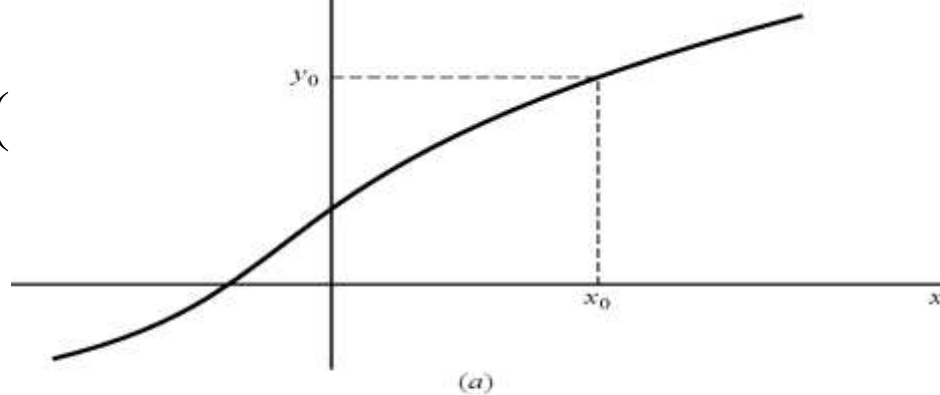
Transformations of a Random Variable

$$Y = T(X)$$

$$f_X(\cdot)$$

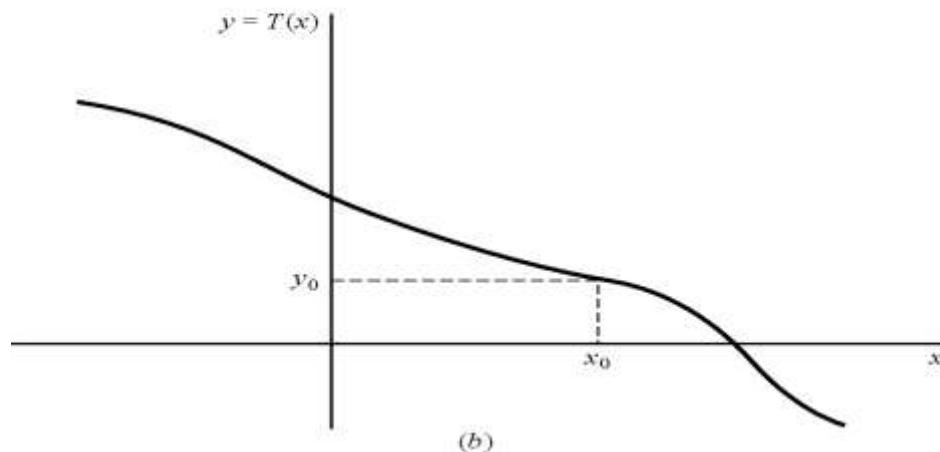
monotone increasing $\Leftrightarrow$

$$T(x_1) < T(x_2) \text{ for any } x_1 < x_2$$



monotone decreasing $\Leftrightarrow$

$$T(x_1) > T(x_2) \text{ for any } x_1 < x_2$$



Assume monotone increasing $T(\bullet)$ $Y = T(X)$

$$F_Y(y_0) = P[Y \leq y_0] = P[X \leq x_0] = F_X(x_0)$$

$$\int_{-\infty}^{y_0} f_Y(y) dy = \int_{-\infty}^{T^{-1}(y_0)} f_X(x) dx$$

$$f_Y(y_0) = f_X[T^{-1}(y_0)] \frac{dT^{-1}(y_0)}{dy_0}$$

$$f_Y(y) = f_X[T^{-1}(y)] \frac{dT^{-1}(y)}{dy} = f_X(x) \frac{dx}{dy}$$

Assume monotone decreasing $T(\bullet)$ $Y = T(X)$

$$F_Y(y_0) = P[Y \leq y_0] = P[X \geq x_0] = 1 - F_X(x_0)$$

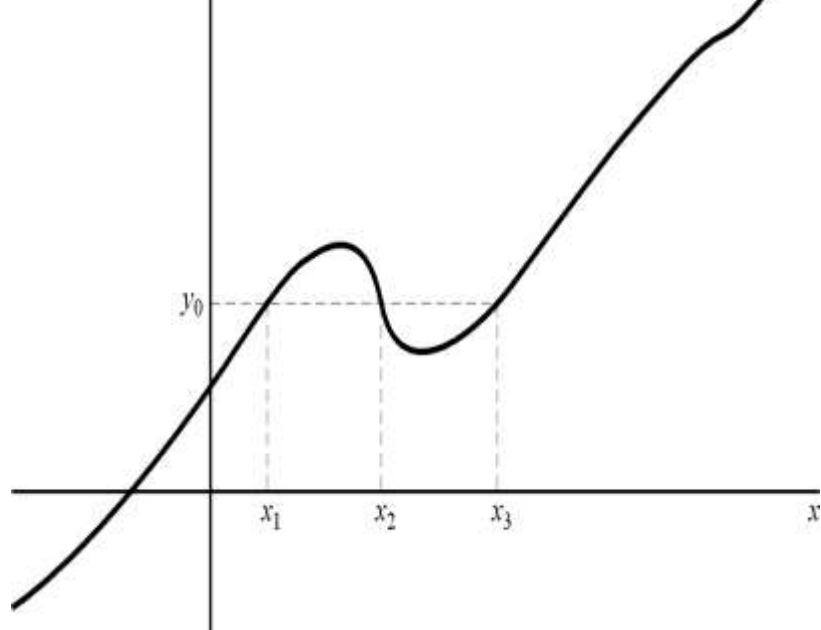
$$f_Y(y) = -f_X(x) \frac{dx}{dy}$$

$$\text{monotone } T(\bullet) \Rightarrow f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right| = f_X(x) \frac{1}{\left| \frac{dy}{dx} \right|}$$

nonmonotone $T(\bullet)$

$$Y = T(X)$$

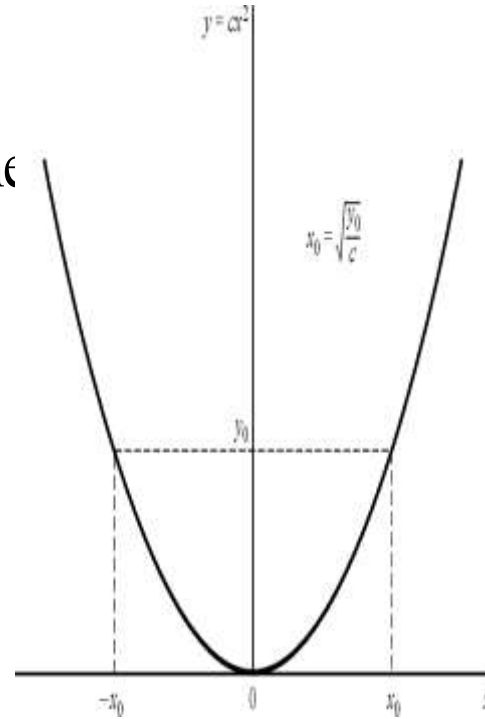
$$f_Y(y) = \sum_n \frac{f_X(x_n)}{\left| \frac{dT(x)}{dx} \right|_{x=x_n}}$$



Ex 3.4-2:

$$Y = T(X) = cX^2 \quad \text{nonmonotone}$$

$$\begin{aligned} f_Y(y) &= f_X(\sqrt{y/c}) \left| \frac{d\sqrt{y/c}}{dy} \right| \\ &\quad + f_X(-\sqrt{y/c}) \left| \frac{-d\sqrt{y/c}}{dy} \right| \\ &= \frac{f_X(\sqrt{y/c}) + f_X(-\sqrt{y/c})}{2\sqrt{cy}}, \quad y \geq 0 \end{aligned}$$



MULTIPLE RANDOM VARIABLES and OPERATIONS:

MULTIPLE RANDOM VARIABLES :

Vector Random Variables

A **vector random variable** X is a function that assigns a vector of real numbers to each outcome ζ in S , the sample space of the random experiment

Events and Probabilities

EXAMPLE 4.4

Consider the tow-dimensional random variable $\mathbf{X} = (X, Y)$. Find the region of the plane corresponding to the events

$$A = \{X + Y \leq 10\},$$

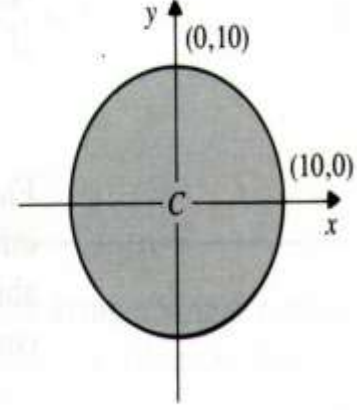
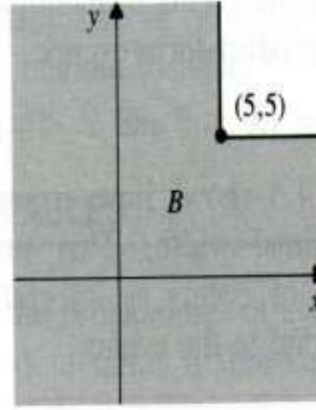
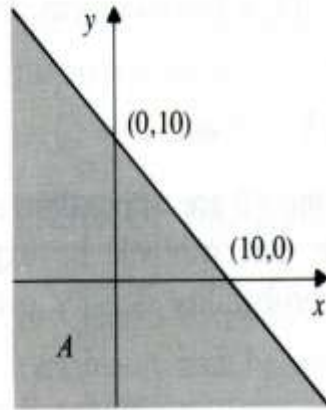
$$B = \{\min(X, Y) \leq 5\}, \text{ and}$$

$$C = \{X^2 + Y^2 \leq 100\}.$$

The regions corresponding to events A and C are straightforward to find and are shown in Fig. 4.1.

FIGURE 4.1

Examples of two-dimensional events.



Independence

If the one-dimensional random variable X and Y are “independent,” if A_1 is any event that involves X only and A_2 is any event that involves Y only, then

$$P[X \text{ in } A_1, Y \text{ in } A_2] = P[X \text{ in } A_1]P[Y \text{ in } A_2].$$

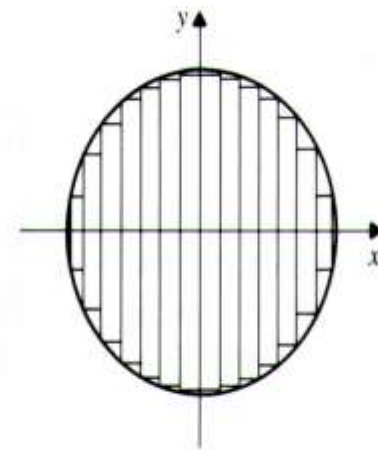
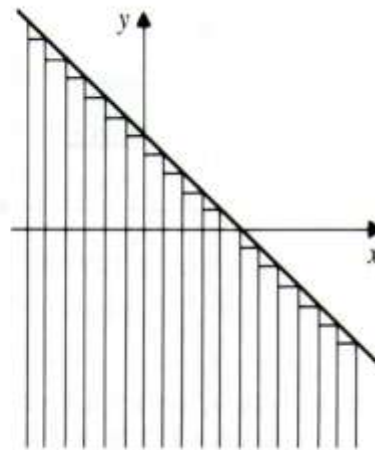
In the general case of n random variables, we say that the random variables $X_1, X_2, \dots, X_n$ are **independent** if

$$P[X_1 \text{ in } A_1, \dots, X_n \text{ in } A_n] = P[X_1 \text{ in } A_1] \cdots P[X_n \text{ in } A_n], \quad (4.3)$$

where the A_k is an event that involves X_k only.

FIGURE 4.3

Some two-dimensional non-product-form events.



Pairs of Discrete Random Variable

Let the vector random variable $\mathbf{X} = (X, Y)$ assume values from some countable set $\mathcal{S} = \{(x_j, y_k), j = 1, 2, \dots, k = 1, 2, \dots\}$. The **joint probability mass function** of $\mathbf{X}$ specifies the probabilities of the product-form event

$$\{X = x_j\} \cap \{Y = y_k\}:$$

$$\begin{aligned} p_{X,Y}(x_j, y_k) &= P[\{X = x_j\} \cap \{Y = y_k\}] \\ &\equiv P[X = x_j, Y = y_k] \quad j = 1, 2, \dots \quad k = 1, 2, \dots \end{aligned} \quad (4.4)$$

The probability of any event A is the sum of the pmf over the outcomes in A

$$P[X \text{ in } A] = \sum_{(x_j, y_k) \text{ in } A} p_{X,Y}(x_j, y_k). \quad (4.5)$$

$$\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} p_{X,Y}(x_j, y_k) = 1. \quad (4.6)$$

The **marginal probability mass functions** :

$$\begin{aligned} p_X(x_j) &= P[X = x_j] \\ &= P[X = x_j, Y = \text{anything}] \\ &= P[\{X = x_j \text{ and } Y = y_1\} \cup \{X = x_j \text{ and } Y = y_2\} \cup \dots] \\ &= \sum_{k=1}^{\infty} p_{X,Y}(x_j, y_k), \end{aligned} \quad (4.7a)$$

$$\begin{aligned} p_Y(y_k) &= P[Y = y_k] \\ &= \sum_{j=1}^{\infty} p_{X,Y}(x_j, y_k). \end{aligned} \quad (4.7b)$$

The Joint cdf of X and Y

The **joint cumulative distribution function of X and Y** is defined as the probability of the product-form event $\{X \leq x_1\} \cap \{Y \leq y_1\}$:

$$F_{X,Y}(x_1, y_1) = P[X \leq x_1, Y \leq y_1]. \quad (4.8)$$

The joint cdf is nondecreasing in the “northeast” direction,

$$(i) \quad F_{X,Y}(x_1, y_1) \leq F_{X,Y}(x_2, y_2) \quad \text{if } x_1 \leq x_2 \text{ and } y_1 \leq y_2,$$

It is impossible for either X or Y to assume a value less than $-\infty$ therefore

$$(ii) \quad F_{X,Y}(-\infty, y_1) = F_{X,Y}(x_2, -\infty) = 0$$

It is certain that X and Y will assume values less than infinity, therefore

$$(iii) \quad F_{X,Y}(\infty, \infty) = 1.$$

If we let one of the variables approach infinity while keeping the other fixed, we obtain the **marginal cumulative distribution functions**

$$(iv) \quad F_X(x) = F_{X,Y}(x, \infty) = P[X \leq x, Y \leq \infty] = P[X \leq x]$$

and

$$F_Y(y) = F_{X,Y}(\infty, y) = P[Y \leq y].$$

Recall that the cdf for a single random variable is continuous from the right. It can be shown that the joint cdf is continuous from the “north” and from the “east”

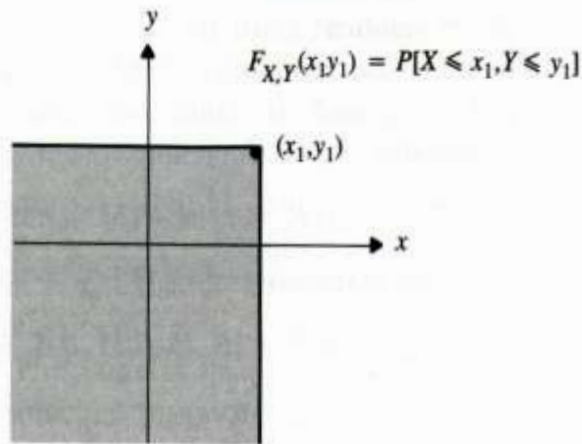
$$(v) \quad \lim_{x \rightarrow a^+} F_{X,Y}(x, y) = F_{X,Y}(a, y)$$

and

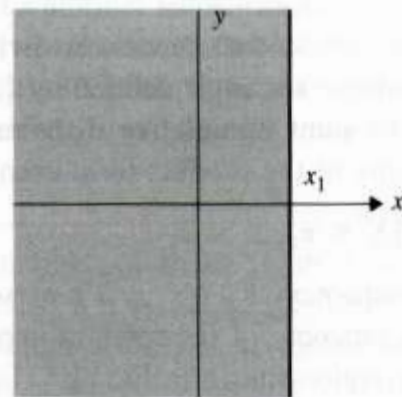
$$\lim_{y \rightarrow b^+} F_{X,Y}(x, y) = F_{X,Y}(x, b)$$

FIGURE 4.4

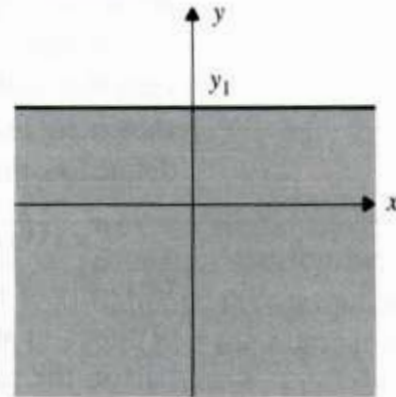
The joint cumulative distribution function is defined as the probability of the semi-infinite rectangle defined by the point (x_1, y_1) .

**FIGURE 4.5**

The marginal cdf's are the probabilities of these half-planes.



$$F_X(x_1) = P[X \leq x_1, Y < \infty]$$



$$F_Y(y_1) = P[X < \infty, Y \leq y_1]$$

The Joint pdf of Two Jointly Continuous Random Variables

We say that the random variables X and Y are jointly continuous if the probabilities of events involving (X, Y) can be expressed as an integral of a pdf. There is a nonnegative function $f_{X,Y}(x,y)$, called the joint probability density function, that is defined on the real plane such that for every event A , a subset of the plane,

$$P[\mathbf{X} \text{ in } A] = \int_A \int f_{X,Y}(x', y') dx' dy', \quad (4.9)$$

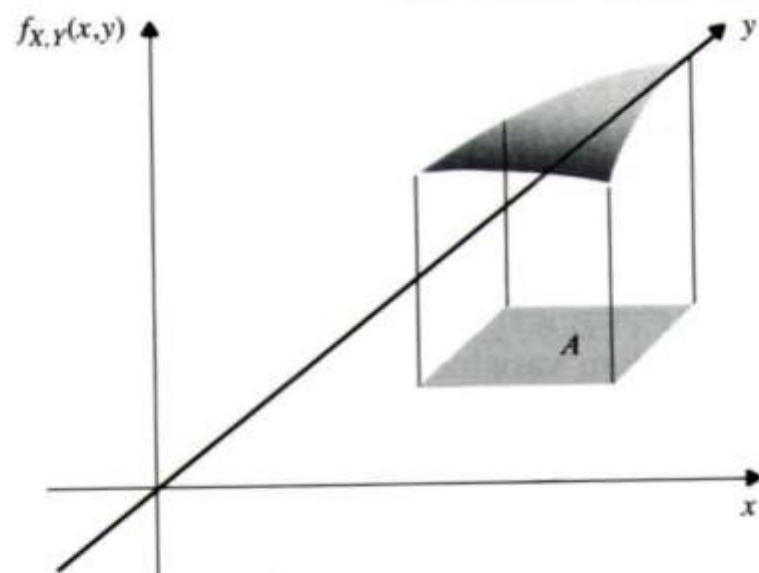
as shown in Fig. 4.7. When A is the entire plane, the integral must equal one :

$$1 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x', y') dx' dy'. \quad (4.10)$$

The joint cdf can be obtained in terms of the joint pdf of jointly continuous random variables by integrating over the semi-infinite

FIGURE 4.7

The probability of A is the integral of $f_{X,Y}(x,y)$ over the region defined by A .



The **marginal pdf's** $f_X(x)$ and $f_Y(y)$ are obtained by taking the derivative of the corresponding marginal cdf's

$$F_X(x) = F_{X,Y}(x, \infty)$$

$$F_Y(y) = F_{X,Y}(\infty, y).$$

$$\begin{aligned} F_X(x) &= \frac{d}{dx} \int_{-\infty}^x \left\{ \int_{-\infty}^{\infty} f_{X,Y}(x', y') dy' \right\} dx' \\ &= \int_{-\infty}^{\infty} f_{X,Y}(x, y') dy'. \end{aligned} \tag{4.15a}$$

$$F_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x', y) dx'. \tag{4.15b}$$

INDEPENDENCE OF TWO RANDOM VARIABLES

X and Y are independent random variables if any event A_1 defined in terms of X is independent of any event A_2 defined in terms of Y ;

$$P[X \text{ in } A_1, Y \text{ in } A_2] = P[X \text{ in } A_1]P[Y \text{ in } A_2]. \quad (4.17)$$

Suppose that X and Y are a pair of discrete random variables. If we let

$A_1 = \{X = x_j\}$ and $A_2 = \{Y = y_k\}$ then the independence of X and Y implies that

$$\begin{aligned} p_{X,Y}(x_j, y_k) &= P[X = x_j, Y = y_k] \\ &= P[X = x_j]P[Y = y_k] \\ &= p_X(x_j)p_Y(y_k) \quad \text{for all } x_j \text{ and } y_k. \end{aligned} \quad (4.18)$$

4.4 CONDITIONAL PROBABILITY AND CONDITIONAL EXPECTATION

Conditional Probability

In Section 2.4, we know

$$P[Y \text{ in } A | X = x] = \frac{P[Y \text{ in } A, X = x]}{P[X = x]}. \quad (4.22)$$

If X is discrete, then Eq. (4.22) can be used to obtain the conditional cdf of Y given $X = x_k$:

$$F_Y(y | x_k) = \frac{P[Y \leq y, X = x_k]}{P[X = x_k]}, \quad \text{for } P[X = x_k] > 0. \quad (4.23)$$

The conditional pdf of Y given $X = x_k$, if the derivative exists, is given by

$$f_Y(y | x_k) = \frac{d}{dy} F_Y(y | x_k). \quad (4.24)$$

MULTIPLE RANDOM VARIABLES

Joint Distributions

The **joint cumulative distribution function** of $X_1, X_2, \dots, X_n$ is defined as the probability of an n -dimensional semi-infinite rectangle associate with the point $(x_1, \dots, x_n)$:

$$F_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = P[X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n]. \quad (4.38)$$

The joint cdf is defined for discrete, continuous, and random variables of mixed type

FUNCTIONS OF SEVERAL RANDOM VARIABLES

One Function of Several Random Variables

Let the random variable Z be defined as a function of several random variables:

$$Z = g(X_1, X_2, \dots, X_n). \quad (4.51)$$

The cdf of Z is found by first finding the equivalent event of that is, the set $R_Z = \{\mathbf{x} = (x_1, \dots, x_n) \text{ such that } g(\mathbf{x}) \leq z\}$, then

$$\begin{aligned} F_Z(z) &= P[\mathbf{X} \text{ in } R_z] \\ &= \int_{\mathbf{x} \text{ in } R_z} \dots \int f_{X_1, \dots, X_n}(x'_1, \dots, x'_n) dx'_1 \dots dx'_n. \end{aligned} \quad (4.52)$$

EXAMPLE 4.31 Sum of Two Random Variables

Let $Z = X + Y$. Find $F_Z(z)$ and $f_Z(z)$ in terms of the joint pdf of X and Y .

The cdf of Z is

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x'} f_{X,Y}(x', y') dy' dx'.$$

The pdf of Z is

$$f_Z(z) = \frac{d}{dz} F_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x', z - x') dx'. \quad (4.53)$$

Thus the pdf for the sum of two random variables is given by a superposition integral. If X and Y are independent random variables, then by Eq. (4.21) the pdf is given by the convolution integral of the marginal pdf's of X and Y :

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x') f_Y(z - x') dx'. \quad (4.54)$$

pdf of Linear Transformations

We consider first the linear transformation of two random variables

$$\begin{aligned} V &= aX + bY \\ W &= cX + eY \end{aligned} \quad \begin{bmatrix} V \\ W \end{bmatrix} = \begin{bmatrix} a & b \\ c & e \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}.$$

Denote the above matrix by A . We will assume A has an inverse, so each point (v, w) has a unique corresponding point (x, y) obtained from

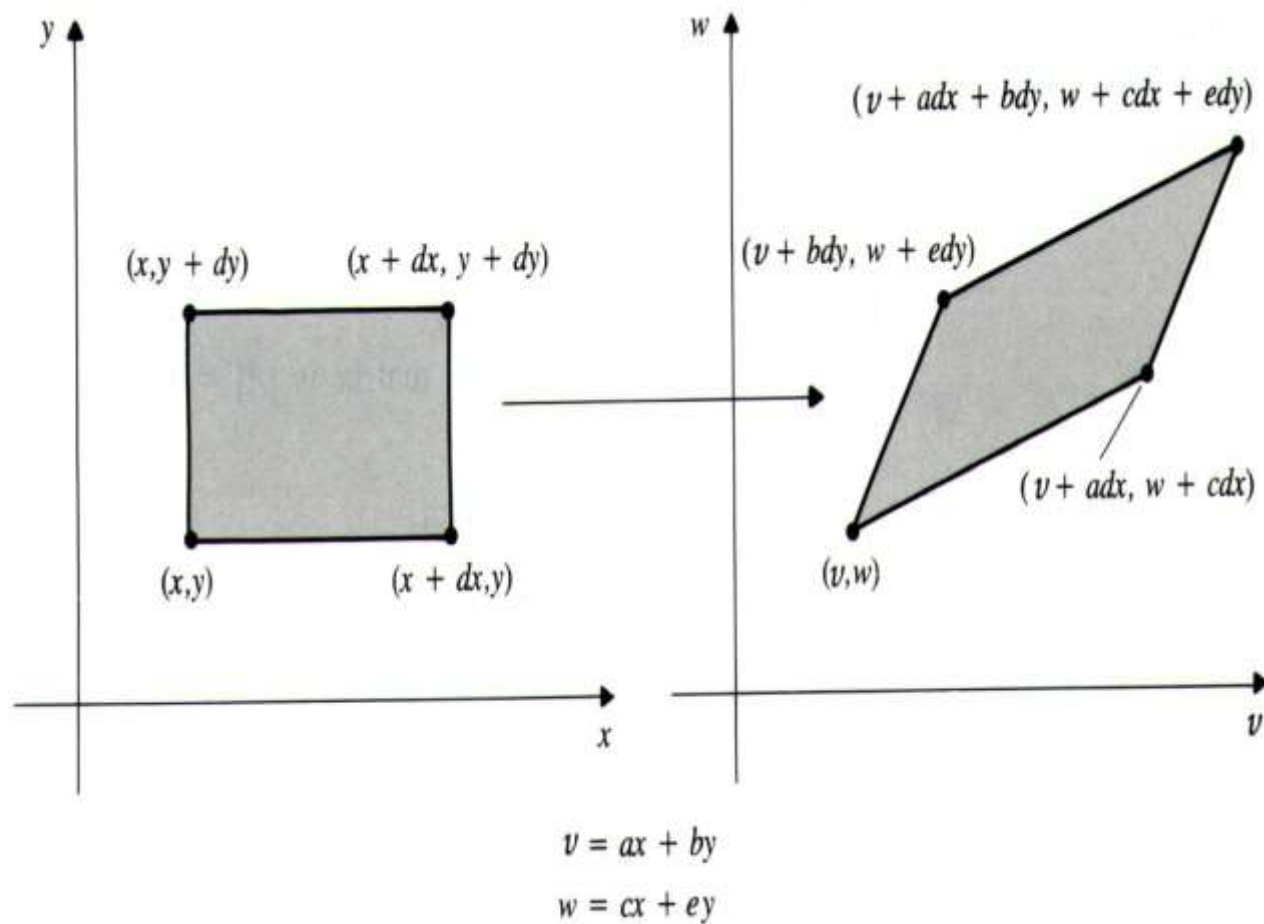
$$\begin{bmatrix} x \\ y \end{bmatrix} = A^{-1} \begin{bmatrix} v \\ w \end{bmatrix}. \quad (4.56)$$

In Fig. 4.15, the infinitesimal rectangle and the parallelogram are equivalent events, so their probabilities must be equal. Thus

$$f_{X,Y}(x, y)dx dy \cong f_{V,W}(v, w)dP$$

FIGURE 4.15

Image of an infinitesimal rectangle under a linear transformation.



where dP is the area of the parallelogram. The joint pdf of V and W is thus given by

$$f_{V,W}(v, w) = \frac{f_{X,Y}(x, y)}{\left| \frac{dP}{dx dy} \right|}, \quad (4.57)$$

where x and y are related to (v, w) by Eq. (4.56)

It can

be shown that $dP = |ae - bc| dx dy$, so the “stretch factor” is

$$\left| \frac{dP}{dx dy} \right| = \frac{|ae - bc|(dx dy)}{(dx dy)} = |ae - bc| = |A|,$$

where $|A|$ is the determinant of A .

Let the n -dimensional vector $\mathbf{Z}$ be

$$\mathbf{Z} = A\mathbf{X},$$

where A is an $n \times n$ invertible matrix. The joint of $\mathbf{Z}$ is then

EXPECTED VALUE OF FUNCTIONS OF RANDOM VARIABLES

The expected value of $Z = g(X, Y)$ can be found using the following expressions

$$E[Z] = \begin{cases} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) & X, Y \text{ jointly continuous} \\ \sum_i \sum_n g(x_i, y_n) p_{X,Y}(x_i, y_n) & X, Y \text{ discrete.} \end{cases} \quad (4.64)$$

*Joint Characteristic Function

The joint characteristic function of n random variables is defined as

$$\Phi_{X_1, X_2, \dots, X_n}(w_1, w_2, \dots, w_n) = E\left[e^{j(w_1 X_1 + w_2 X_2 + \dots + w_n X_n)}\right]. \quad (4.73a)$$

$$\Phi_{X,Y}(w_1, w_2) = E\left[e^{j(w_1 X + w_2 Y)}\right]. \quad (4.73b)$$

If X and Y are jointly continuous random variables, then

$$\Phi_{X,Y}(w_1, w_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) e^{j(w_1 x + w_2 y)} dx dy. \quad (4.73c)$$

The inversion formula for the Fourier transform implies that the joint pdf is given by

$$f_{X,Y}(x, y) = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \Phi_{X,Y}(w_1, w_2) e^{j(w_1 x + w_2 y)} dw_1 dw_2. \quad (4.74)$$

JOINTLY GAUSSIAN RANDOM VARIABLES

The random variables X and Y are said to be jointly Gaussian if their joint pdf has the form

$$f_{X,Y}(x,y) = \frac{\exp\left\{\frac{-1}{2(1-\rho_{X,Y}^2)}\left[\left(\frac{x-m_1}{\sigma_1}\right)^2 - 2\rho_{X,Y}\left(\frac{x-m_1}{\sigma_1}\right)\left(\frac{y-m_2}{\sigma_2}\right) + \left(\frac{y-m_2}{\sigma_2}\right)^2\right]\right\}}{2\pi\sigma_1\sigma_2\sqrt{1-\rho_{X,Y}^2}} \quad (4.79)$$

$$-\infty < x < \infty \quad \text{and} \quad -\infty < y < \infty$$

The pdf is constant for values x and y for which the argument of the exponent is constant

$$\left[\left(\frac{x - m_1}{\sigma_1} \right)^2 - 2\rho_{X,Y} \left(\frac{x - m_1}{\sigma_1} \right) \left(\frac{y - m_2}{\sigma_2} \right) + \left(\frac{y - m_2}{\sigma_2} \right)^2 \right] = \text{constant}$$

When $\rho_{X,Y} = 0$, X and Y are independent ; when $\rho_{X,Y} \neq 0$, the major axis of the ellipse is oriented along the angle

$$\theta = \frac{1}{2} \arctan \left(\frac{2\rho_{X,Y}\sigma_1\sigma_2}{\sigma_1^2 - \sigma_2^2} \right). \quad (4.80)$$

Note that the angle is 45° when the variance are equal.

The marginal pdf of X is found by integrating $f_{X,Y}(x, y)$ over all y

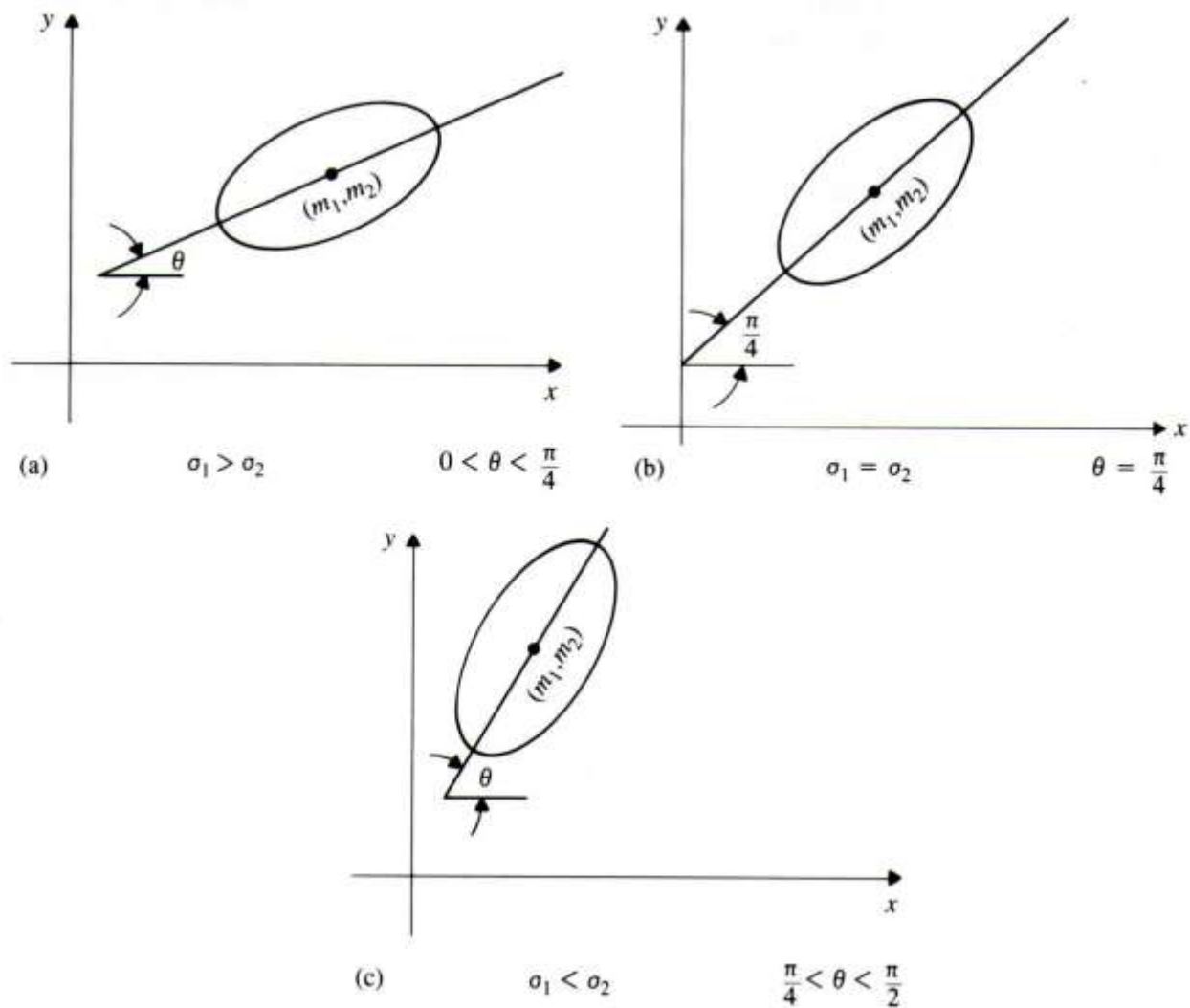
$$f_X(x) = \frac{e^{-(x-m_1)^2 / 2\sigma_1^2}}{\sqrt{2\pi}\sigma_1}, \quad (4.81)$$

that is, X is a Gaussian random variable with mean m_1 and variance

$$\sigma_1^2$$

FIGURE 4.19

Orientation of contours of equal value of joint Gaussian pdf for $\rho_{X,Y} > 0$.



n Jointly Gaussian Random Variables

The random variables $X_1, X_2, \dots, X_n$ are said to be jointly Gaussian if their joint pdf is given by

$$f_{\mathbf{x}}(\mathbf{x}) \equiv f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = \frac{\exp\left\{-\frac{1}{2}(\mathbf{x} - \mathbf{m})^T K^{-1}(\mathbf{x} - \mathbf{m})\right\}}{(2\pi)^{n/2} |K|^{1/2}}, \quad (4.83)$$

where $\mathbf{x}$ and $\mathbf{m}$ are column vectors defined by

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{m} = \begin{bmatrix} m_1 \\ m_2 \\ \vdots \\ m_n \end{bmatrix} = \begin{bmatrix} E[X_1] \\ E[X_2] \\ E[X_3] \\ E[X_4] \end{bmatrix}$$

and K is the **covariance matrix** that is defined by

$$K = \begin{bmatrix} \text{VAR}(X_1) & \text{COV}(X_2, X_1) & \dots & \text{COV}(X_1, X_n) \\ \text{COV}(X_2, X_1) & \text{VAR}(X_2) & \dots & \text{COV}(X_2, X_n) \\ \vdots & \vdots & & \vdots \\ \text{COV}(X_n, X_1) & \dots & & \text{VAR}(X_n) \end{bmatrix} \quad (4.84)$$

Transformations of Random Vectors

Let $X_1, \dots, X_n$ be random variables associate with some experiment, and let the random variables $Z_1, \dots, Z_n$ be defined by n functions of $\mathbf{X} = (X_1, \dots, X_n)$:

$$Z_1 = g_1(\mathbf{X}) \quad Z_2 = g_2(\mathbf{X}) \quad \dots \quad Z_n = g_n(\mathbf{X}).$$

The joint cdf of $Z_1, \dots, Z_n$ at the point $\mathbf{z} = (z_1, \dots, z_n)$ is equal to the probability of the region of $\mathbf{x}$ where

$$F_{Z_1, \dots, Z_n}(z_1, \dots, z_n) = P[g_1(\mathbf{X}) \leq z_1, \dots, g_n(\mathbf{X}) \leq z_n]. \quad (4.55a)$$

$$F_{Z_1, \dots, Z_n}(z_1, \dots, z_n) = \int \cdots \int_{\mathbf{x}': g_k(\mathbf{x}') \leq z_k} f_{X_1, \dots, X_n}(x'_1, \dots, x'_n) dx'_1 \cdots dx'_n. \quad (4.55b)$$

pdf of Linear Transformations

We consider first the linear transformation of two random variables

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Denote the above matrix by A . We will assume A has an inverse, so each point (v, w) has a unique corresponding point (x, y) obtained from

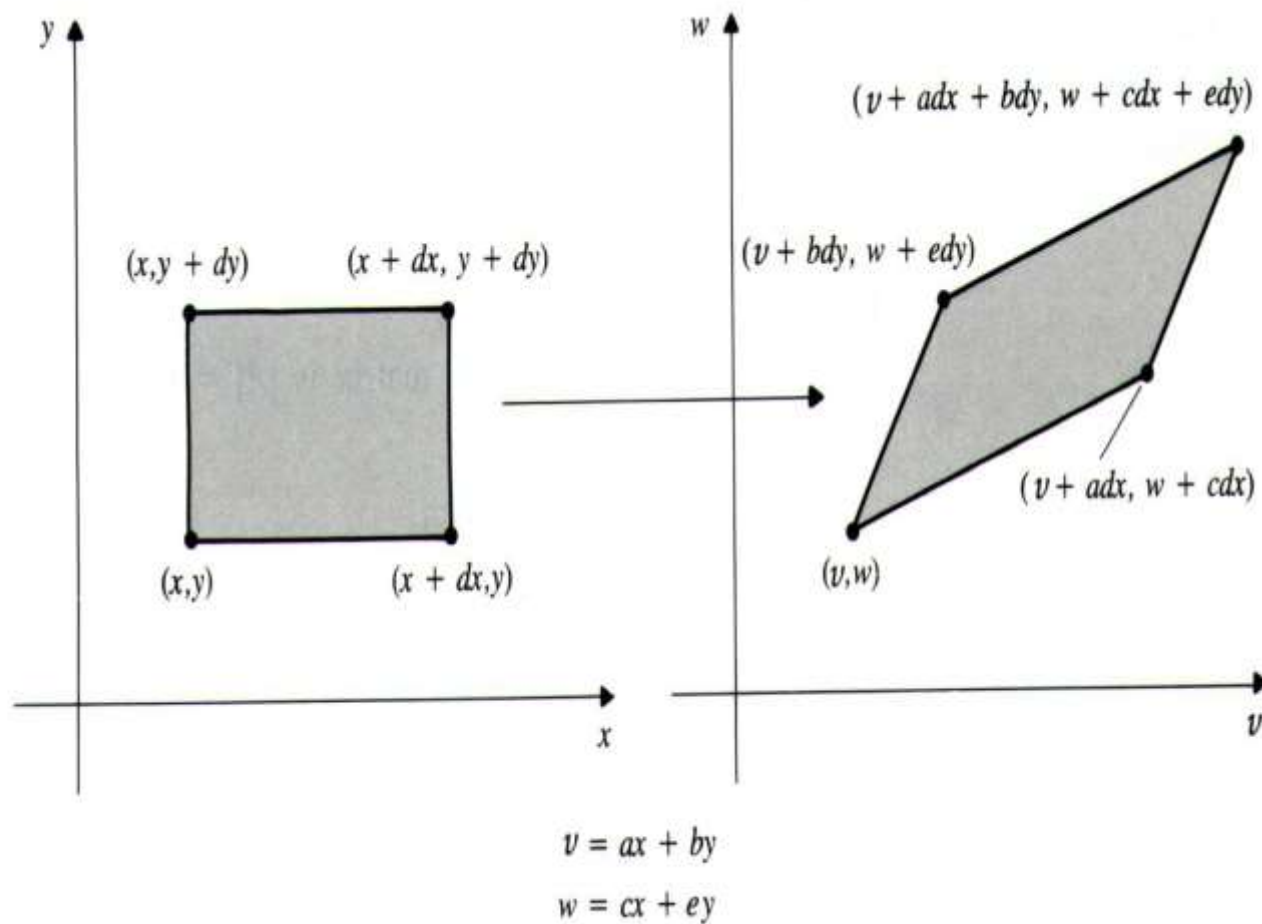
$$\begin{bmatrix} x \\ y \end{bmatrix} = A^{-1} \begin{bmatrix} v \\ w \end{bmatrix}. \tag{4.56}$$

In Fig. 4.15, the infinitesimal rectangle and the parallelogram are equivalent events, so their probabilities must be equal. Thus

$$f_{X,Y}(x, y)dx dy \cong f_{V,W}(v, w)dP$$

FIGURE 4.15

Image of an infinitesimal rectangle under a linear transformation.



Stochastic Processes

Let ξ denote the random outcome of an experiment. To every such outcome suppose a waveform $X(t, \xi)$ is assigned.

The collection of such waveforms form a stochastic process. The set of $\{\xi_k\}$ and the time index t can be continuous or discrete (countably infinite or finite) as well.

For fixed $\xi_i \in S$ (the set of all experimental outcomes), $X(t, \xi_i)$ is a specific time function.

For fixed t ,

$$X_1 = X(t_1, \xi_i)$$

is a random variable. The ensemble of all such realizations $X(t, \xi)$ over time represents the stochastic

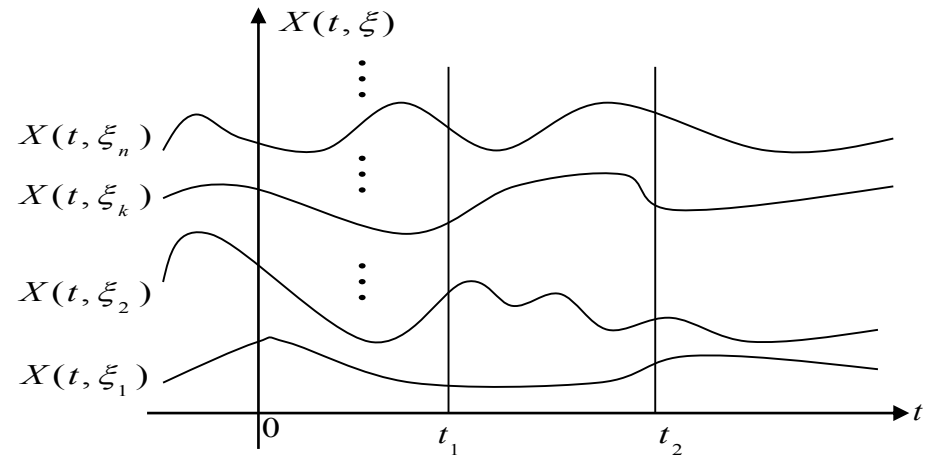


Fig. 14.1

process $X(t)$. (see Fig 14.1). For example

$$X(t) = a \cos(\omega_0 t + \varphi),$$

If $X(t)$ is a stochastic process, then for fixed t , $X(t)$ represents a random variable. Its distribution function is given by

$$F_x(x, t) = P\{X(t) \leq x\}$$

Notice that $F_x(x, t)$ depends on t , since for a different t , we obtain a different random variable. Further

$$f_x(x, t) = \frac{dF_x(x, t)}{dx}$$

represents the first-order probability density function of the process $X(t)$.

For $t = t_1$ and $t = t_2$, $X(t)$ represents two different random variables $X_1 = X(t_1)$ and $X_2 = X(t_2)$ respectively. Their joint distribution is given by

$$F_x(x_1, x_2, t_1, t_2) = P\{X(t_1) \leq x_1, X(t_2) \leq x_2\}$$

and

$$f_x(x_1, x_2, t_1, t_2) = \frac{\partial^2 F_x(x_1, x_2, t_1, t_2)}{\partial x_1 \partial x_2}$$

represents the second-order density function of the process $X(t)$. Similarly $f_x(x_1, x_2, \dots, x_n, t_1, t_2, \dots, t_n)$ represents the n^{th} order density function of the process $X(t)$. Complete specification of the stochastic process $X(t)$ requires the knowledge of $f_x(x_1, x_2, \dots, x_n, t_1, t_2, \dots, t_n)$ for all i , $i = 1, 2, \dots, n$ and for all n . (an almost impossible task in reality).

Mean of a Stochastic Process:

$$\mu(t) = E\{X(t)\} = \int_{-\infty}^{+\infty} x f_X(x, t) dx$$

represents the mean value of a process $X(t)$. In general, the mean of a process can depend on the time index t .

Autocorrelation function of a process $X(t)$ is defined as

$$R_{XX}(t_1, t_2) = E\{X(t_1)X^*(t_2)\} = \int \int x_1 x_2^* f_X(x_1, x_2, t_1, t_2) dx_1 dx_2$$

and it represents the interrelationship between the random variables $X_1 = X(t_1)$ and $X_2 = X(t_2)$ generated from the process $X(t)$.

Properties:

1. $R_{XX}(t_1, t_2) = R_{XX}^*(t_2, t_1) = [E\{X(t_2)X^*(t_1)\}]^*$
2. $R_{XX}(t, t) = E\{|X(t)|^2\} > 0.$

3. $R_{xx}(t_1, t_2)$ represents a nonnegative definite function, i.e., for *any* set of constants $\{a_i\}_{i=1}^n$

$$\sum_{i=1}^n \sum_{j=1}^n a_i a_j^* R_{xx}(t_i, t_j) \geq 0. \quad (14-8)$$

Eq. (14-8) follows by noticing that

The function

$$E\{|Y|^2\} \geq 0 \quad \text{for } Y = \sum_{i=1}^n a_i X(t_i).$$

represents the **autocovariance** function of the process $X(t)$.

(14-9)

Example 14.1

Let

$$C_{xx}(t_1, t_2) = R_{xx}(t_1, t_2) - \mu_x(t_1)\mu_x^*(t_2)$$

Then

$$z = \int_{-T}^T X(t) dt.$$

$$\begin{aligned} E[|z|^2] &= \int_{-T}^T \int_{-T}^T E\{X(t_1)X^*(t_2)\} dt_1 dt_2 \\ &= \int_{-T}^T \int_{-T}^T R_{xx}(t_1, t_2) dt_1 dt_2 \end{aligned} \quad (14-10)$$

Stationary Stochastic Processes

Stationary processes exhibit statistical properties that are invariant to shift in the time index. Thus, for example, second-order stationarity implies that the statistical properties of the pairs $\{X(t_1), X(t_2)\}$ and $\{X(t_1+c), X(t_2+c)\}$ are the same for *any* c . Similarly first-order stationarity implies that the statistical properties of $X(t_i)$ and $X(t_i+c)$ are the same for any c .

In strict terms, the statistical properties are governed by the joint probability density function. Hence a process is n^{th} -order **Strict-Sense Stationary (S.S.S)** if

$$f_x(x_1, x_2, \dots, x_n, t_1, t_2, \dots, t_n) \equiv f_x(x_1, x_2, \dots, x_n, t_1 + c, t_2 + c, \dots, t_n + c)$$

for *any* c , where the left side represents the joint density function of the random variables $X_1 = X(t_1), X_2 = X(t_2), \dots, X_n = X(t_n)$ and the right side corresponds to the joint density function of the random variables $X'_1 = X(t_1 + c), X'_2 = X(t_2 + c), \dots, X'_n = X(t_n + c)$.

A process $X(t)$ is said to be **strict-sense stationary** if (14-14) is true for all $t_i, i = 1, 2, \dots, n, n = 1, 2, \dots$ and *any* c .

For a **first-order strict sense stationary process**,
from (14-14) we have

$$f_x(x, t) \equiv f_x(x, t + c) \quad (14-15)$$

for any c . In particular $c = -t$ gives

$$f_x(x, t) = f_x(x) \quad (14-16)$$

i.e., the first-order density of $X(t)$ is independent of t . In that case

Similarly, for a **second-order strict-sense stationary process**
we have from (14-14)

$$E[X(t)] = \int_{-\infty}^{+\infty} x f(x) dx = \mu, \text{ a constant.} \quad (14-17)$$

for any c . For $c = -t_2$ we get

$$f_x(x_1, x_2, t_1, t_2) \equiv f_x(x_1, x_2, t_1 + c, t_2 + c)$$

$$f_x(x_1, x_2, t_1, t_2) \equiv f_x(x_1, x_2, t_1 - t_2) \quad (14-18)$$

i.e., the second order density function of a strict sense stationary process depends only on the difference of the time indices

In that case the autocorrelation function is given by

$$t_1 - t_2 = \tau.$$

$$\begin{aligned} R_{xx}(t_1, t_2) &= E\{X(t_1)X^*(t_2)\} \\ &\triangleq \int \int x_1 x_2^* f_x(x_1, x_2, \tau = t_1 - t_2) dx_1 dx_2 \\ &= R_{xx}(t_1 - t_2) = R_{xx}(\tau) = R_{xx}^*(-\tau), \end{aligned} \quad (14-19)$$

i.e., the autocorrelation function of a second order strict-sense stationary process depends only on the difference of the time indices $\tau = t_1 - t_2$.

Notice that (14-17) and (14-19) are consequences of the stochastic process being first and second-order strict sense stationary.

On the other hand, the basic conditions for the first and second order stationarity – Eqs. (14-16) and (14-18) – are usually difficult to verify.

In that case, we often resort to a looser definition of stationarity, known as **Wide-Sense Stationarity (W.S.S)**, by making use of

(14-17) and (14-19) as the necessary conditions. Thus, a process $X(t)$ is said to be **Wide-Sense Stationary** if

(i) $E\{X(t)\} = \mu$ (14-20)

(ii) $E\{X(t_1)X^*(t_2)\} = R_{xx}(t_1 - t_2),$ (14-21)

i.e., for wide-sense stationary processes, the mean is a constant and the autocorrelation function depends only on the difference between the time indices. Notice that (14-20)-(14-21) does not say anything about the nature of the probability density functions, and instead deal with the average behavior of the process. Since (14-20)-(14-21) follow from (14-16) and (14-18), strict-sense stationarity always implies wide-sense stationarity. However, the converse is *not true* in general, the only exception being the Gaussian process.

This follows, since if $X(t)$ is a Gaussian process, then by definition

$X_1 = X(t_1), X_2 = X(t_2), \dots, X_n = X(t_n)$ are jointly Gaussian random variables for any $t_1, t_2, \dots, t_n$ whose joint characteristic function is given by

$$\phi_{\underline{X}}(\omega_1, \omega_2, \dots, \omega_n) = e^{j \sum_{k=1}^n \mu(t_k) \omega_k - \sum_{l,k}^n \sum_{l,k} C_{xx}(t_l, t_k) \omega_l \omega_k / 2} \quad (14-22)$$

where $C_{xx}(t_i, t_k)$ is as defined on (14-9). If $X(t)$ is wide-sense stationary, then using (14-20)-(14-21) in (14-22) we get

$$\phi_{\underline{X}}(\omega_1, \omega_2, \dots, \omega_n) = e^{j \sum_{k=1}^n \mu \omega_k - \frac{1}{2} \sum_{l=1}^n \sum_{k=1}^n C_{xx}(t_l - t_k) \omega_l \omega_k} \quad (14-23)$$

and hence if the set of time indices are shifted by a constant c to generate a new set of jointly Gaussian random variables $X'_1 = X(t_1 + c)$, $X'_2 = X(t_2 + c), \dots, X'_n = X(t_n + c)$ then their joint characteristic function is identical to (14-23). Thus the set of random variables and $\{X'_i\}_{i=1}^n$ have the same joint probability distribution for all n and $\{X_i\}_{i=1}^n$ all c , establishing the strict sense stationarity of Gaussian processes from its wide-sense stationarity.

To summarize if $X(t)$ is a Gaussian process, then
wide-sense stationarity (w.s.s) $\Rightarrow$ strict-sense stationarity (s.s.s).
Notice that since the joint p.d.f of Gaussian random variables depends only on their second order statistics, which is also the basis

Systems with Stochastic Inputs

A deterministic system¹ transforms each input waveform $X(t, \xi_i)$ into an output waveform $Y(t, \xi_i) = T[X(t, \xi_i)]$ by operating only on the time variable t . Thus a set of realizations at the input corresponding to a process $X(t)$ generates a new set of realizations $\{Y(t, \xi)\}$ at the output associated with a new process $Y(t)$.

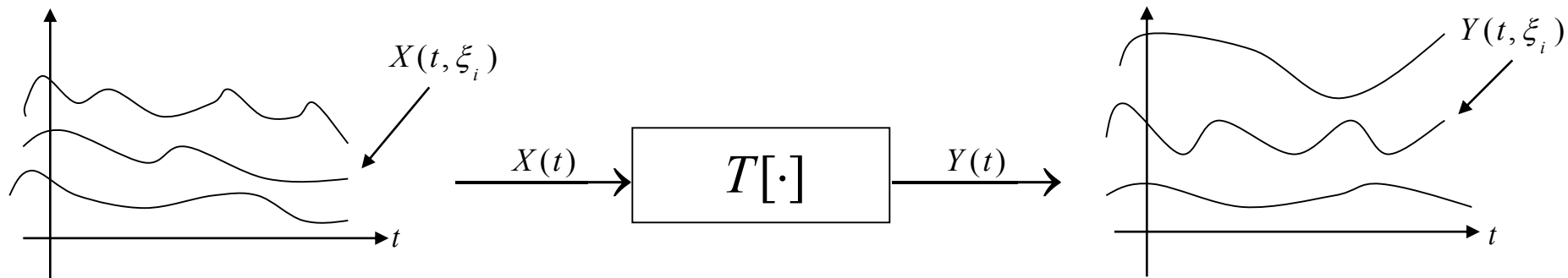


Fig. 14.3

Our goal is to study the output process statistics in terms of the input process statistics and the system function.

¹A stochastic system on the other hand operates on both the variables t and ξ .

Linear Systems: $L[\cdot]$ represents a linear system if

$$L\{a_1 X(t_1) + a_2 X(t_2)\} = a_1 L\{X(t_1)\} + a_2 L\{X(t_2)\}. \quad (14-28)$$

Let

$$Y(t) = L\{X(t)\} \quad (14-29)$$

represent the output of a linear system.

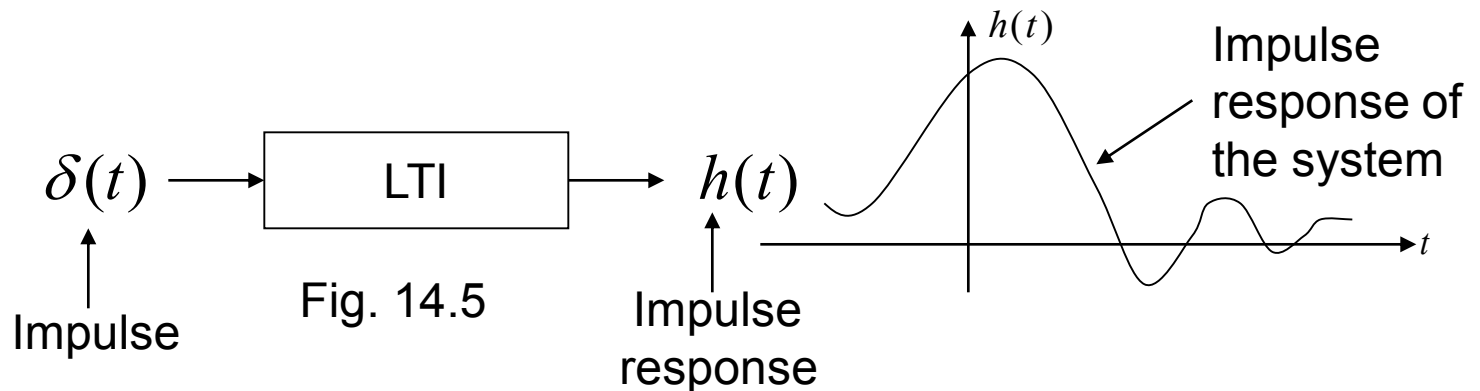
Time-Invariant System: $L[\cdot]$ represents a time-invariant system if

$$Y(t) = L\{X(t)\} \Rightarrow L\{X(t - t_0)\} = Y(t - t_0)$$

i.e., shift in the input results in the same shift in the output also.

If $L[\cdot]$ satisfies both (14-28) and (14-30), then it corresponds to a linear time-invariant (LTI) system. (14-30)

LTI systems can be uniquely represented in terms of their output to a delta function



then

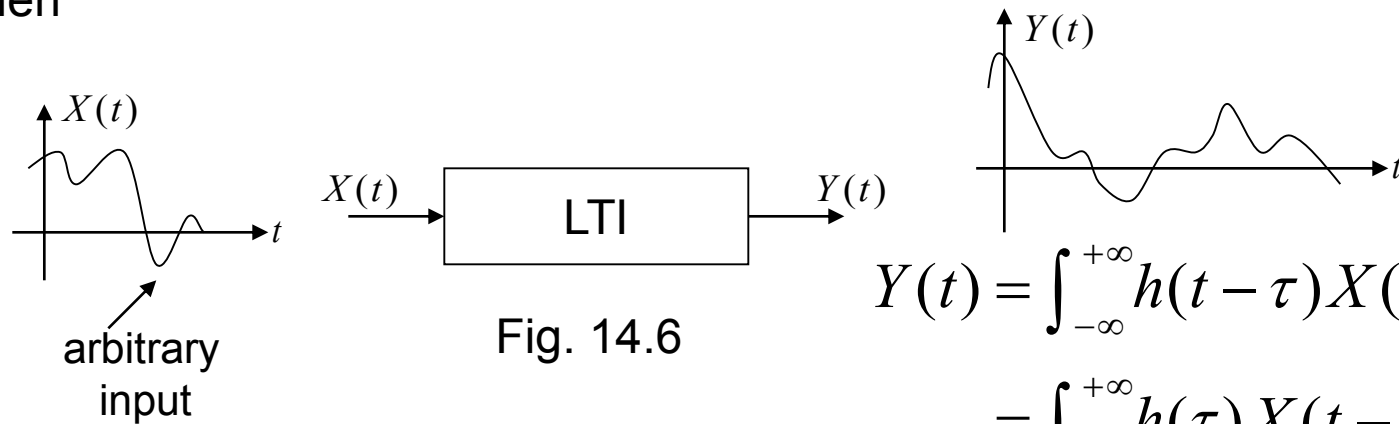


Fig. 14.6

$$Y(t) = \int_{-\infty}^{+\infty} h(t - \tau) X(\tau) d\tau$$

$$= \int_{-\infty}^{+\infty} h(\tau) X(t - \tau) d\tau \quad (14-31)$$

Eq. (14-31) follows by expressing $X(t)$ as

$$X(t) = \int_{-\infty}^{+\infty} X(\tau) \delta(t - \tau) d\tau \quad (14-32)$$

and applying (14-28) and (14-30) to $Y(t) = L\{X(t)\}$ Thus

$$Y(t) = L\{X(t)\} = L\left\{\int_{-\infty}^{+\infty} X(\tau) \delta(t - \tau) d\tau\right\}$$

$$= \int_{-\infty}^{+\infty} L\{X(\tau) \delta(t - \tau)\} d\tau \quad \leftarrow \text{By Linearity}$$

$$= \int_{-\infty}^{+\infty} X(\tau) L\{\delta(t - \tau)\} d\tau \quad \leftarrow \text{By Time-invariance}$$

$$= \int_{-\infty}^{+\infty} X(\tau) h(t - \tau) d\tau = \int_{-\infty}^{+\infty} h(\tau) X(t - \tau) d\tau. \quad (14-33)$$

Output Statistics: Using (14-33), the mean of the output process is given by

$$\begin{aligned}\mu_Y(t) &= E\{Y(t)\} = \int_{-\infty}^{+\infty} E\{X(\tau)h(t-\tau)d\tau\} \\ &= \int_{-\infty}^{+\infty} \mu_X(\tau)h(t-\tau)d\tau = \mu_X(t) * h(t).\end{aligned}\tag{14-34}$$

Similarly the cross-correlation function between the input and output processes is given by

$$\begin{aligned}R_{XY}(t_1, t_2) &= E\{X(t_1)Y^*(t_2)\} \\ &= E\{X(t_1)\int_{-\infty}^{+\infty} X^*(t_2 - \alpha)h^*(\alpha)d\alpha\} \\ &= \int_{-\infty}^{+\infty} E\{X(t_1)X^*(t_2 - \alpha)\}h^*(\alpha)d\alpha \\ &= \int_{-\infty}^{+\infty} R_{XX}(t_1, t_2 - \alpha)h^*(\alpha)d\alpha \\ &= R_{XX}(t_1, t_2) * h^*(t_2).\end{aligned}\tag{14-35}$$

Finally the output autocorrelation function is given by

$$\begin{aligned}
R_{YY}(t_1, t_2) &= E\{Y(t_1)Y^*(t_2)\} \\
&= E\left\{\int_{-\infty}^{+\infty} X(t_1 - \beta)h(\beta)d\beta Y^*(t_2)\right\} \\
&= \int_{-\infty}^{+\infty} E\{X(t_1 - \beta)Y^*(t_2)\}h(\beta)d\beta \\
&= \int_{-\infty}^{+\infty} R_{XY}(t_1 - \beta, t_2)h(\beta)d\beta \\
&= R_{XY}(t_1, t_2) * h(t_1),
\end{aligned} \tag{14-36}$$

or

$$R_{YY}(t_1, t_2) = R_{XX}(t_1, t_2) * h^*(t_2) * h(t_1). \tag{14-37}$$

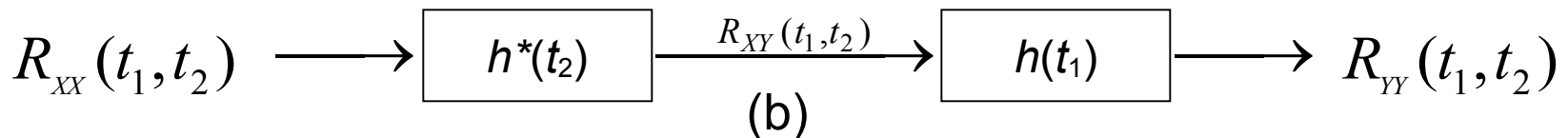
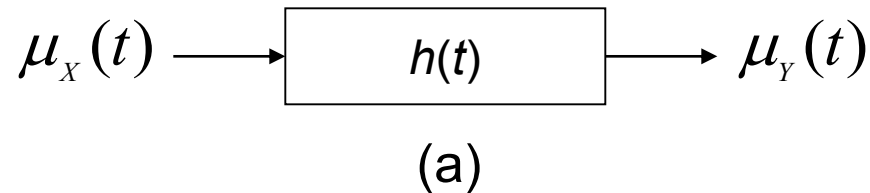


Fig. 14.7

In particular if $X(t)$ is wide-sense stationary, then we have so that from (14-34)

$$\mu_X(t) = \mu_X$$

$$\mu_Y(t) = \mu_X \int_{-\infty}^{+\infty} h(\tau) d\tau = \mu_X c, \quad a \text{ constant.} \quad (14-38)$$

Also $R_{XX}(t_1, t_2) = R_{XX}(t_1 - t_2)$ so that (14-35) reduces to

$$\begin{aligned} R_{XY}(t_1, t_2) &= \int_{-\infty}^{+\infty} R_{XX}(t_1 - t_2 + \alpha) h^*(\alpha) d\alpha \\ &= R_{XX}(\tau) * h^*(-\tau) = R_{XY}(\tau), \quad \tau = t_1 - t_2. \end{aligned} \quad (14-39)$$

Thus $X(t)$ and $Y(t)$ are jointly w.s.s. Further, from (14-36), the output autocorrelation simplifies to

$$\begin{aligned} R_{YY}(t_1, t_2) &\triangleq \int_{-\infty}^{+\infty} R_{XY}(t_1 - \beta - t_2) h(\beta) d\beta, \quad \tau = t_1 - t_2 \\ &= R_{XY}(\tau) * h(\tau) = R_{YY}(\tau). \end{aligned} \quad (14-40)$$

From (14-37), we obtain

$$R_{YY}(\tau) = R_{XX}(\tau) * h^*(-\tau) * h(\tau). \quad (14-41)$$

From (14-38)-(14-40), the output process is also wide-sense stationary.
This gives rise to the following representation

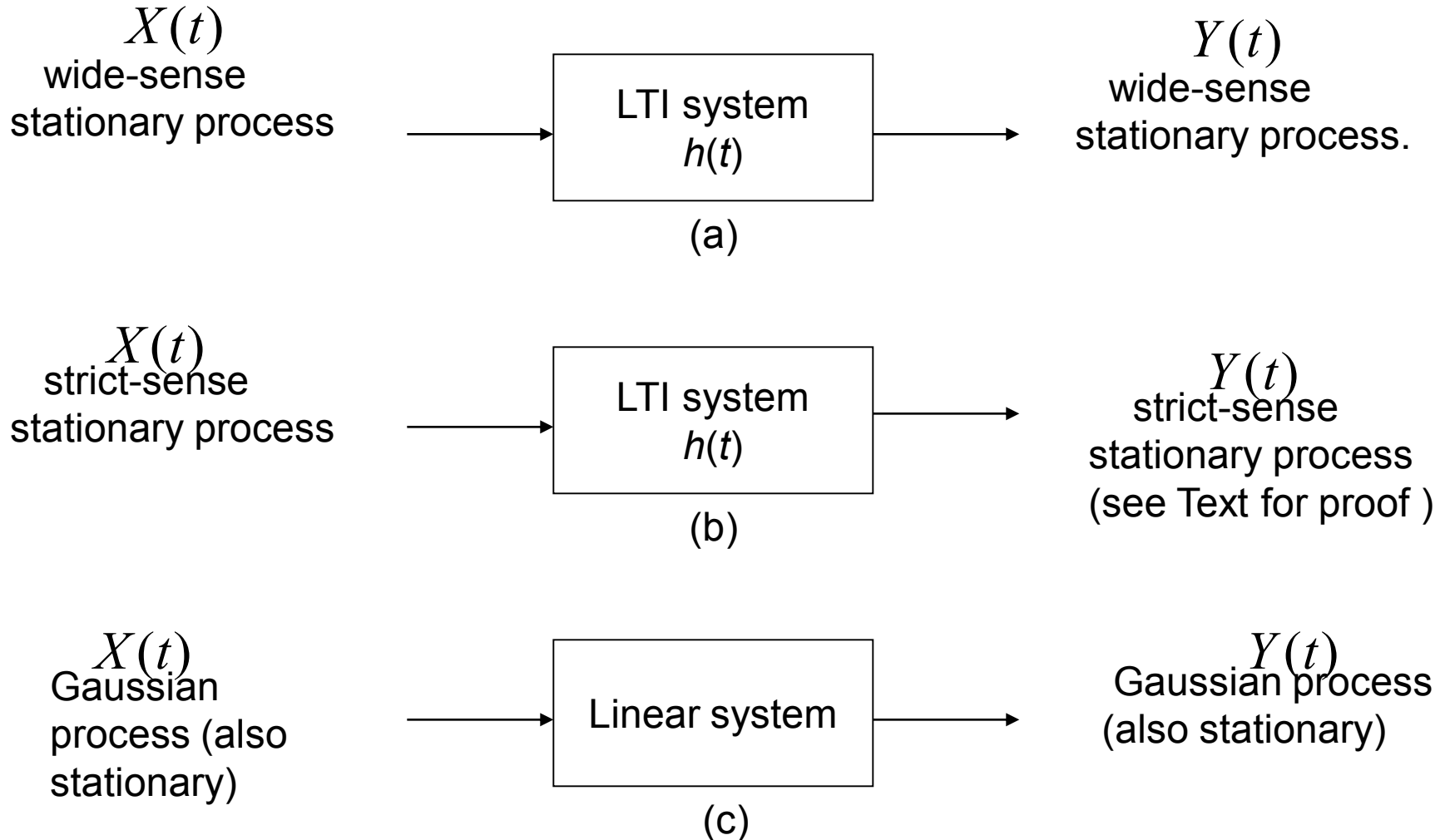


Fig. 14.8

Discrete Time Stochastic Processes:

A discrete time stochastic process $X_n = X(nT)$ is a sequence of random variables. The mean, autocorrelation and auto-covariance functions of a discrete-time process are gives by

$$\mu_n = E\{X(nT)\} \quad (14-57)$$

and

$$R(n_1, n_2) = E\{X(n_1T)X^*(n_2T)\} \quad (14-58)$$

respectively. As before strict sense stationarity and wide-sense stationarity definitions apply here also.

For example, $X(nT)$ is wide sense stationary if

$$C(n_1, n_2) = R(n_1, n_2) - \mu_{n_1} \mu_{n_2}^* \quad (14-59)$$

and

$$E\{X(nT)\} = \mu, \quad a \text{ constant} \quad (14-60)$$

$$E[X\{(k+n)T\}X^*\{(k)T\}] = R(n) = r_n \triangleq r_{-n}^* \quad (14-61)$$

Power Spectrum

For a deterministic signal $x(t)$, the spectrum is well defined: If $X(\omega)$ represents its Fourier transform, i.e., if

$$X(\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt, \quad (18-1)$$

then $|X(\omega)|^2$ represents its energy spectrum. This follows from Parseval's theorem since the signal energy is given by

$$\int_{-\infty}^{+\infty} x^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(\omega)|^2 d\omega = E. \quad (18-2)$$

Thus $|X(\omega)|^2 \Delta\omega$ represents the signal energy in the band $(\omega, \omega + \Delta\omega)$ (see Fig 18.1).

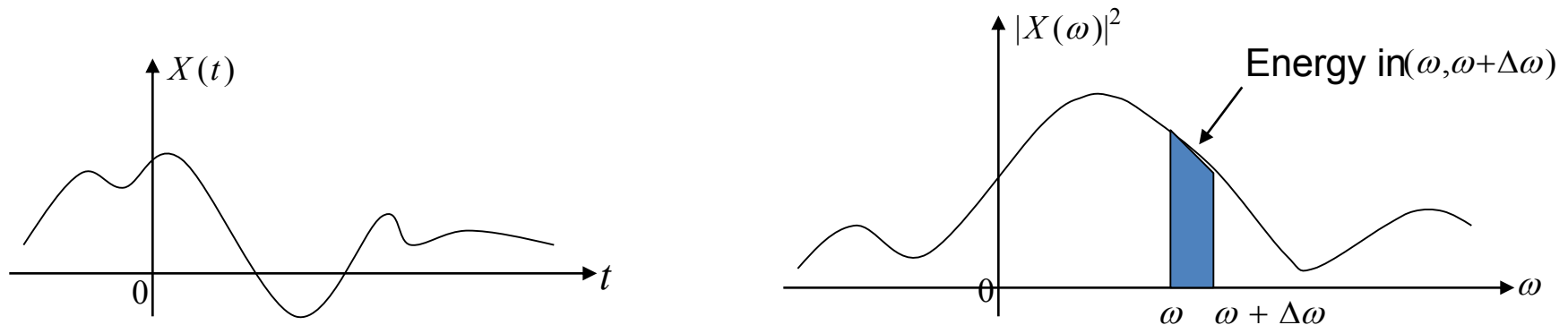


Fig 18.1

However for stochastic processes, a direct application of (18-1) generates a sequence of random variables for every ω . Moreover, for a stochastic process, $E\{|X(t)|^2\}$ represents the ensemble average power (instantaneous energy) at the instant t .

To obtain the spectral distribution of power versus frequency for stochastic processes, it is best to avoid infinite intervals to begin with, and start with a finite interval $(-T, T)$ in (18-1). Formally, partial Fourier transform of a process $X(t)$ based on $(-T, T)$ is given by

so that

$$X_T(\omega) = \int_{-T}^T X(t)e^{-j\omega t} dt \quad (18-3)$$

represents the power distribution associated with that realization based on $(-T, T)$. Notice that (18-4) represents a random variable for every ω and its ensemble average gives, the average power distribution based on $(-T, T)$. Thus

$$\frac{|X_T(\omega)|^2}{2T} = \frac{1}{2T} \left| \int_{-T}^T X(t)e^{-j\omega t} dt \right|^2 \quad \omega, \quad (18-4)$$

$$\begin{aligned}
 P_T(\omega) &= E \left\{ \frac{|X_T(\omega)|^2}{2T} \right\} = \frac{1}{2T} \int_{-T}^T \int_{-T}^T E \{ X(t_1) X^*(t_2) \} e^{-j\omega(t_1-t_2)} dt_1 dt_2 \\
 &= \frac{1}{2T} \int_{-T}^T \int_{-T}^T R_{xx}(t_1, t_2) e^{-j\omega(t_1-t_2)} dt_1 dt_2
 \end{aligned} \tag{18-5}$$

represents the power distribution of $X(t)$ based on $(-T, T)$. For wide sense stationary (w.s.s) processes, it is possible to further simplify (18-5). Thus if $X(t)$ is assumed to be w.s.s, then and (18-5) simplifies to

$$R_{xx}(t_1, t_2) = R_{xx}(t_1 - t_2)$$

Let $\tau = t_1 - t_2$ and proceeding as in (14-24), we get

$$P_T(\omega) = \frac{1}{2T} \int_{-T}^T \int_{-T}^T R_{xx}(t_1 - t_2) e^{-j\omega(t_1-t_2)} dt_1 dt_2.$$

to be the power distribution of the w.s.s. process $X(t)$ based on $(-T, T)$. Finally letting $T \rightarrow \infty$ in (18-6), we obtain

$$\begin{aligned}
 P_T(\omega) &= \frac{1}{2T} \int_{-2T}^{2T} R_{xx}(\tau) e^{-j\omega\tau} (2T - |\tau|) d\tau \\
 &= \int_{-2T}^{2T} R_{xx}(\tau) e^{-j\omega\tau} \left(1 - \frac{|\tau|}{2T}\right) d\tau \geq 0
 \end{aligned} \tag{18-6}$$

$$S_{XX}(\omega) = \lim_{T \rightarrow \infty} P_T(\omega) = \int_{-\infty}^{+\infty} R_{XX}(\tau) e^{-j\omega\tau} d\tau \geq 0 \quad (18-7)$$

to be the *power spectral density* of the w.s.s process $X(t)$. Notice that

$$R_{XX}(\omega) \xleftrightarrow{\text{F.T.}} S_{XX}(\omega) \geq 0. \quad (18-8)$$

i.e., the autocorrelation function and the power spectrum of a w.s.s Process form a Fourier transform pair, a relation known as the **Wiener-Khinchin Theorem**. From (18-8), the inverse formula gives

and in particular for $\tau = 0$, we get

$$R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} S_{XX}(\omega) e^{j\omega\tau} d\omega \quad (18-9)$$

From (18-10), the area under $S_{XX}(\omega)$ represents the total power of the process $X(t)$, and hence $S_{XX}(\omega)$ truly represents the power spectrum. (Fig 18.2).

$$\frac{1}{2\pi} \int_{-\infty}^{+\infty} S_{XX}(\omega) d\omega = R_{XX}(0) = E\{|X(t)|^2\} = P, \quad \text{the total power} \quad (18-10)$$

If $X(t)$ is a real w.s.s process, then $R_{XX}(\tau) = R_{XX}(-\tau)$ so that

$$\begin{aligned} S_{XX}(\omega) &= \int_{-\infty}^{+\infty} R_{XX}(\tau) e^{-j\omega\tau} d\tau \\ &= \int_{-\infty}^{+\infty} R_{XX}(\tau) \cos \omega\tau d\tau \\ &= 2 \int_0^{\infty} R_{XX}(\tau) \cos \omega\tau d\tau = S_{XX}(-\omega) \geq 0 \end{aligned}$$

so that the power spectrum is an even function, (in addition to being real and nonnegative). (18-13)